

Complete JAVASCRIPT

Course Notes

Contains 1 PDF lessons

Generated: 11 September 2026

KnowledgeGate

Module

JavaScript

Topic

JavaScript

Subtopic

JavaScript

Lesson

JavaScript_Notes

Lesson 1 of 1

COMPLETE JAVASCRIPT

14 HOURS



**MYNTRA
PROJECT**



CERTIFICATE

NOTES

You  Tube

[Video Link](#)



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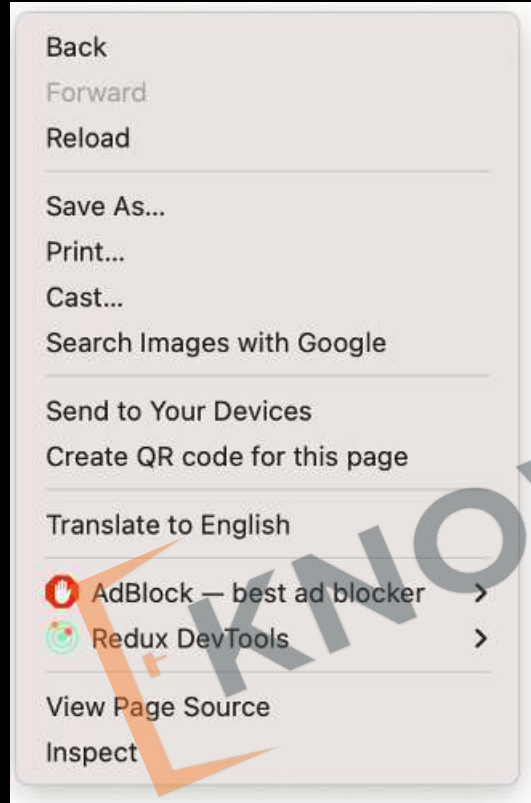


[Sanchit Socket](#)

Introduction to JavaScript

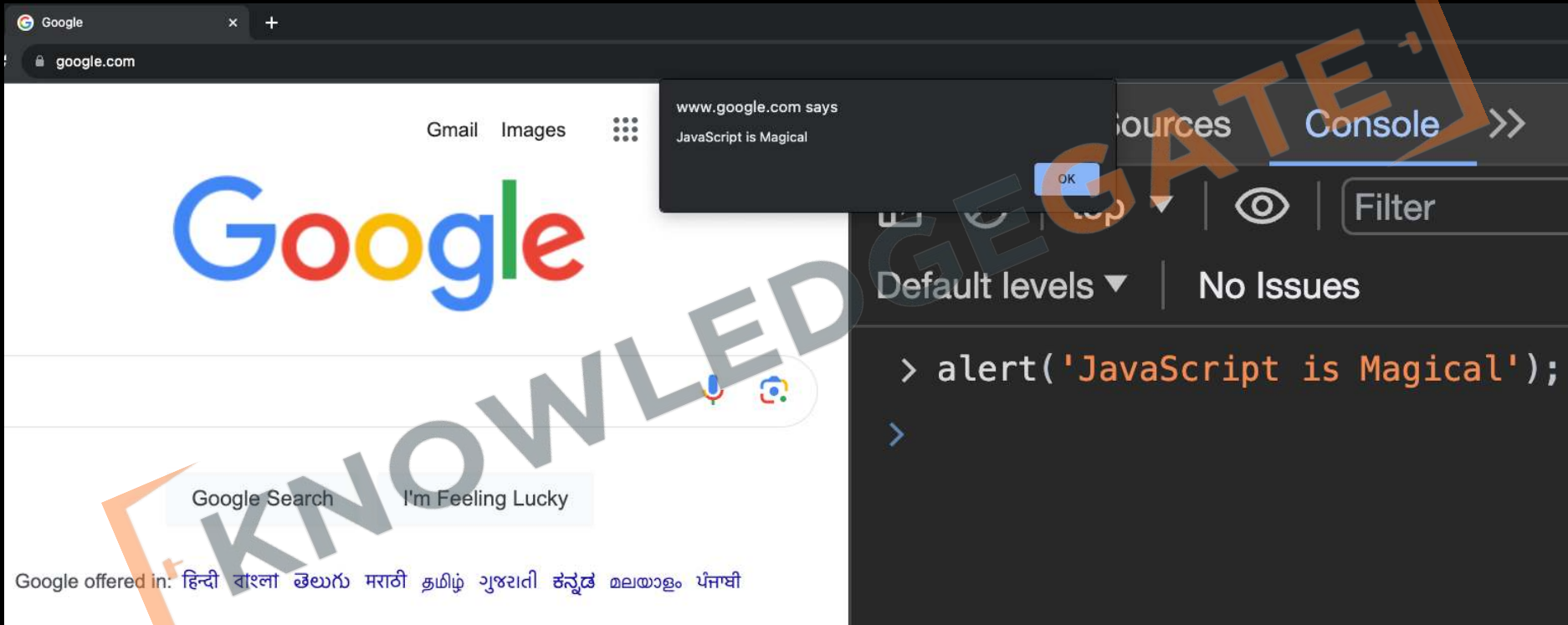
1. JS in Console
2. DOM Manipulation
3. Chrome Extensions
4. What is a Programming Language?
5. What is Syntax?
6. HTML/CSS/JavaScript

1. JS in Console (Inspect)



1. Allows **real-time editing** of **HTML/CSS/JS**
 2. **Run Scripts**: Test code in console.
 3. **Debug**: Locate and fix errors.
 4. **Modify DOM**: Change webpage elements.
- Errors: View error messages.

1. JS in Console (Alert)

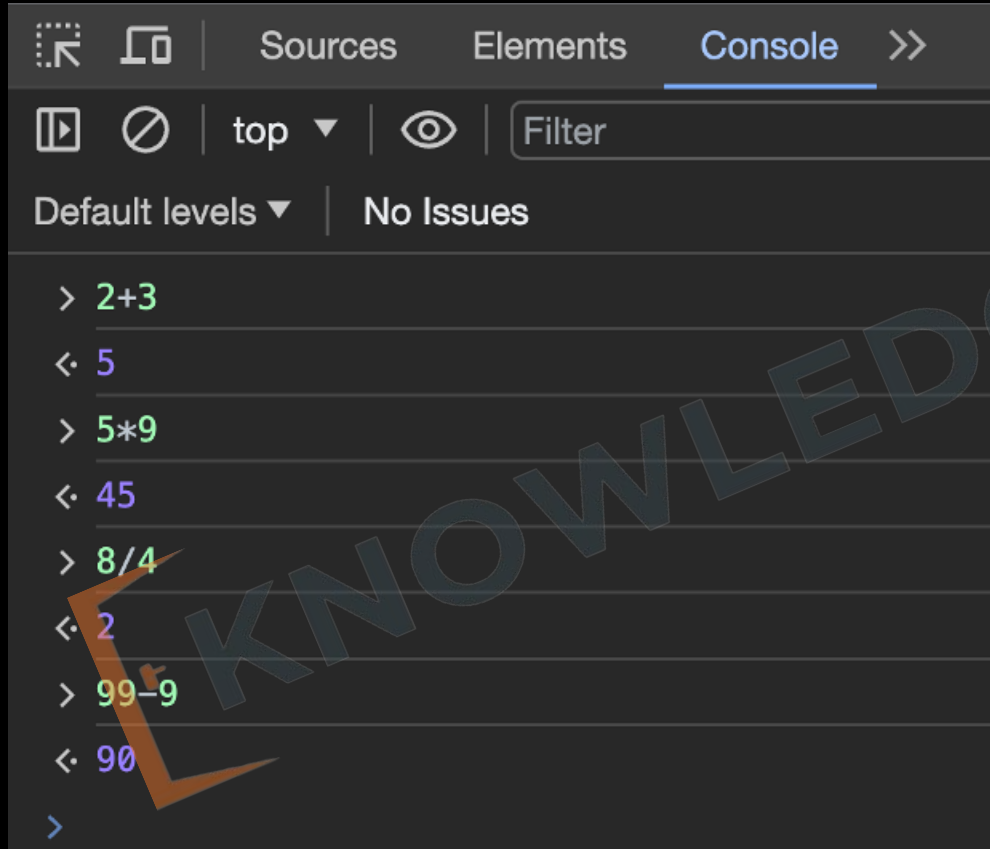


The image shows a screenshot of the Google homepage in a web browser. The browser's address bar shows "google.com". The Google logo is prominently displayed in the center. Below the logo are the "Google Search" and "I'm Feeling Lucky" buttons. At the bottom, there is a line of text in Hindi: "Google offered in: हिन्दी बांग्ला తెలుగు मराठी தமிழ் ગુજરાતી ಕನ್ನಡ മലയാളം ਪੰਜਾਬੀ".

On the right side of the browser window, the "Console" tab is open, showing a message from "www.google.com" that says "JavaScript is Magical". The message is displayed in a dark box with a blue "OK" button. Below the message, the console shows the code snippet: `> alert('JavaScript is Magical');` followed by a prompt character `>`.

A large, semi-transparent watermark "KNOWLEDGEGATE" is overlaid diagonally across the entire image.

1. JS in Console (Math)



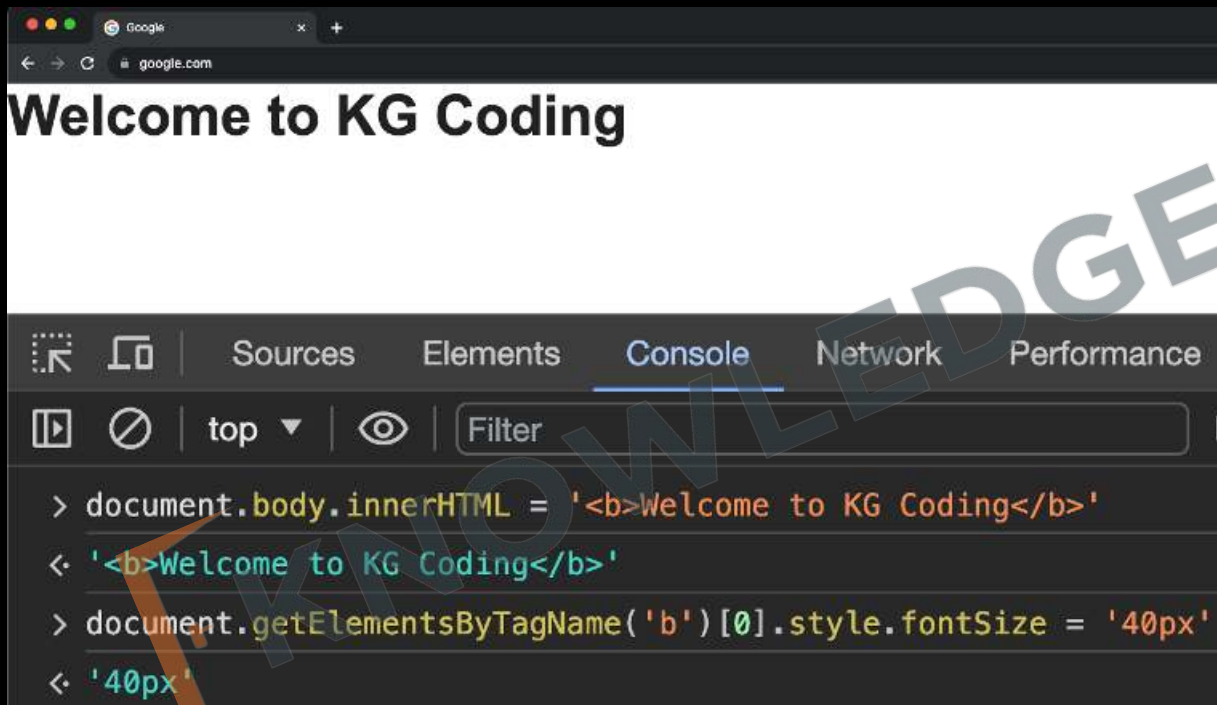
The screenshot shows a web browser's developer console with the 'Console' tab selected. The console displays a series of JavaScript math operations and their results. The input expressions are shown in green, and the results are shown in purple. The operations include addition, multiplication, division, and subtraction. A large, semi-transparent watermark reading 'KNOWLEDGEGATE' is overlaid diagonally across the console area.

```
> 2+3
< 5
> 5*9
< 45
> 8/4
< 2
> 99-9
< 90
>
```

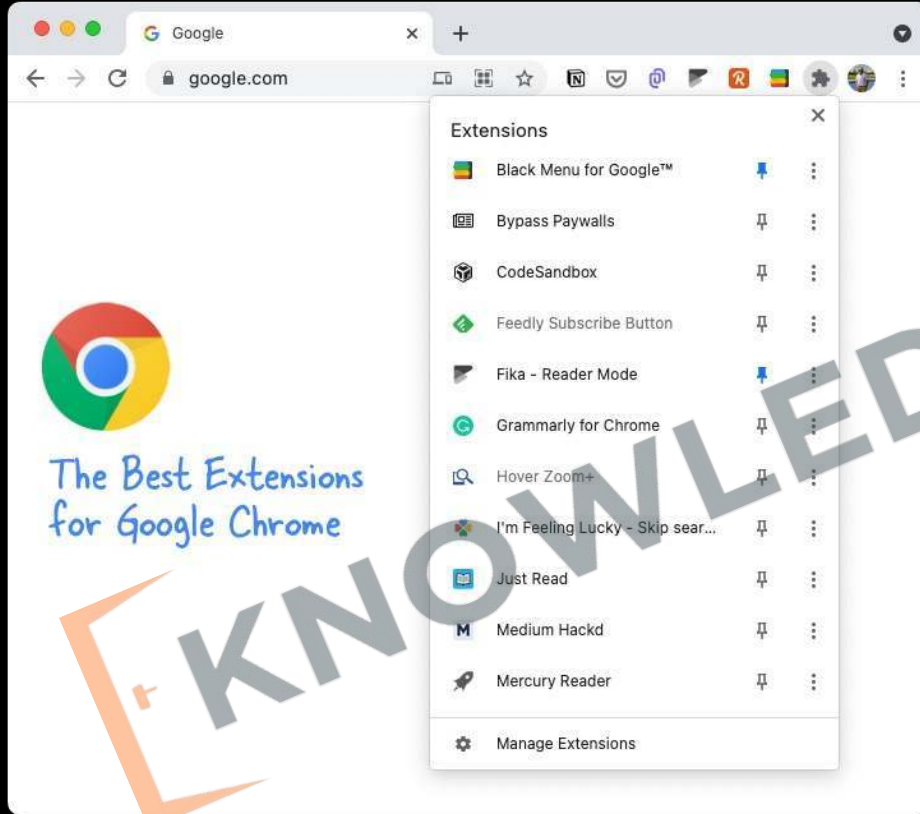
Console can be used
as a Calculator

2. DOM Manipulation

1. Change HTML
2. Change CSS
3. Perform Actions

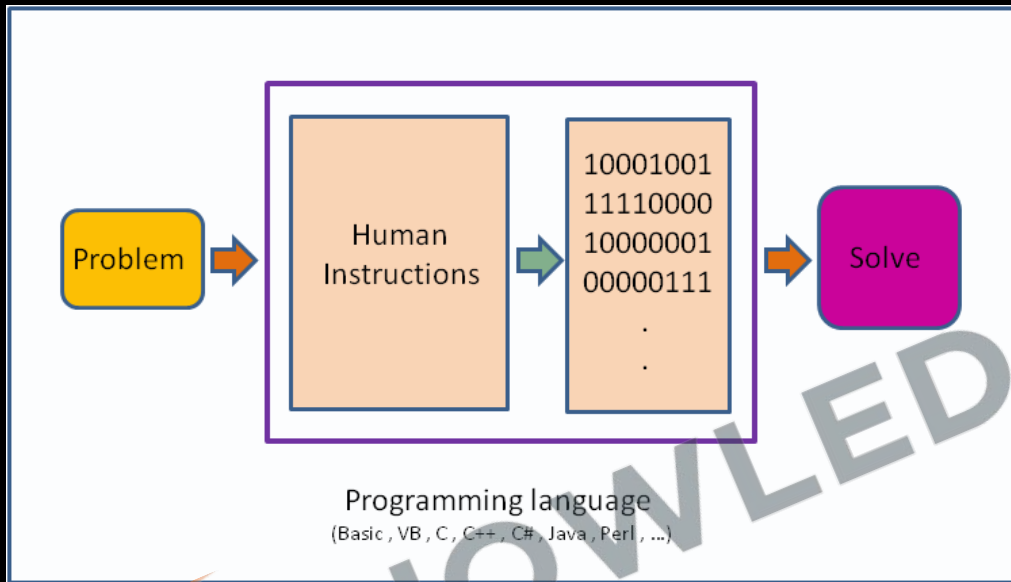


3. Chrome Extensions



1. **Create Features:** Add new functionalities to Chrome.
2. **Interact with Web:** Modify or read webpage content.
3. **API Access:** Use Chrome's built-in functions.
4. **User Experience:** Enhance or customize browsing.

4. What is a Programming Language?



- Giving instructions to a computer
- **Instructions:** Tells computer what to do.
- These instructions are called **code**.

5. What is a Syntax?



- Structure of words in a sentence.
- Rules of the language.
- For programming **exact** syntax must be followed.

```
> alert 'hello world'
```

```
✖ Uncaught SyntaxError: Unexpected string
```

```
> |
```

6. FrontEnd / BackEnd / FullStack



**Client Side / Front-End
Web Development**



**Server Side
Back-End**



Full Stack

6. Role of Browser



1. **Displays Web Page:** Turns HTML code into what you see on screen.
2. **User Clicks:** Helps you interact with the web page.
3. **Updates Content:** Allows changes to the page using JavaScript.
4. **Loads Files:** Gets HTML, images, etc., from the server.



6. HTML

(Hypertext Markup Language)

1. **Structure:** Sets up the layout.
2. **Content:** Adds text, images, links.
3. **Tags:** Uses elements like `<p>`, `<a>`.
4. **Hierarchy:** Organizes elements in a tree.



HTML is required for JavaScript

COMPLETE

 HTML

4 HOUR

PROJECT

CERTIFICATE

CODE



NOTES



Ex- amazon  Microsoft



6. CSS

(Cascading Style Sheets)

1. **Style:** Sets the look and feel.
2. **Colors & Fonts:** Customizes text and background.
3. **Layout:** Controls position and size.
4. **Selectors:** Targets specific **HTML** elements.



CSS is required for JavaScript

COMPLETE



CSS

7 HOUR



MYNTRA
PROJECT



CERTIFICATE

NOTES



Ex-amazon



Microsoft



6. JS

(Java Script)

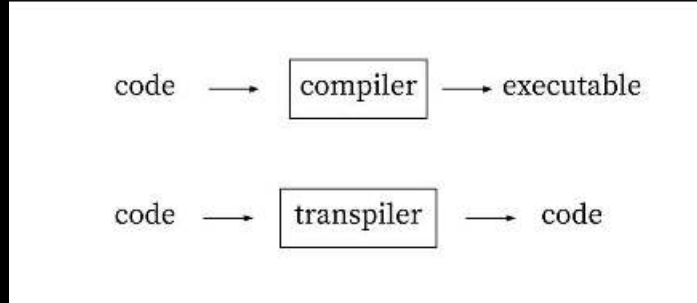
1. JavaScript has nothing to do with Java
2. **Actions:** Enables interactivity.
3. **Updates:** Alters page without reloading.
4. **Events:** **Responds** to user actions.
5. **Data:** Fetches and sends info to server.



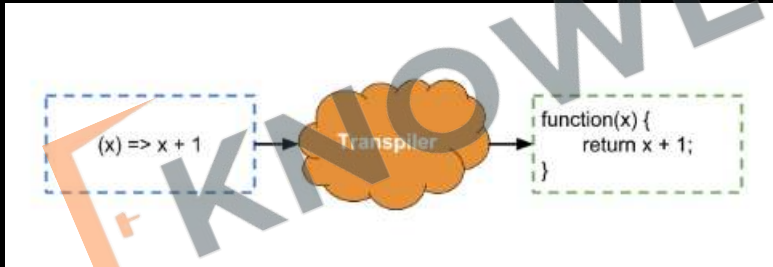


6. JS

(Java Script)

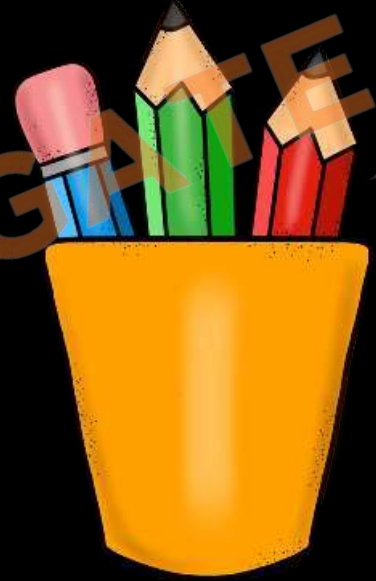


1. JavaScript runs at the **client side** in the browser.
2. Coffee Script / **TypeScript** are **transpiled** to **JavaScript**.



Introduction Revision

1. JS in Console
2. DOM Manipulation
3. Chrome Extensions
4. What is a Programming Language?
5. What is Syntax?
6. HTML/CSS/JavaScript



Practice Exercise

Introduction

1. Use an **alert** to display Good Morning.
2. **Display** your name in a **popup**.
3. Using Math calculate the following:
=> **75-25**
=> **3+3-5**
4. Change **Facebook** page to display **"I am Learning JS"**



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- [Complete HTML](#)
- [Complete CSS](#)
- [Complete JavaScript](#)
- [Complete React and Redux](#)
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Numbers & Strings

7. Arithmetic Operators
8. Order of Operations
9. Types of Numbers (Integers & Floats)
10. Don't learn syntax.
11. Strings
12. TypeOf Operator

7. Arithmetic Operators

Operators	Meaning	Example	Result
+	Addition	4+2	6
-	Subtraction	4-2	2
*	Multiplication	4*2	8
/	Division	4/2	2
%	Modulus operator to get remainder in integer division	5%2	1

MYNTRA Cart

✓ 2/2 ITEMS SELECTED

REMOVE

MOVE TO WISHLIST



Allen Solly

Men Textured Slim Fit Mid-Rise Trousers

Sold by: Westbury Holdings Pvt Ltd_Madura

Size: 32 - Qty: 1 -

₹ 1,474 ~~₹2,499~~ 41% OFF

Coupon Discount: ₹29

🕒 14 days return available

✓ Delivery by 8 Oct 2023



GUESS

Brand Logo Woven Design Structured Handheld Bag

Sold by: Supercom Net

Size: Onesize - Qty: 1 - 9 left

₹ 14,039 ~~₹15,599~~ 10% OFF

Coupon Discount: ₹272

🕒 7 days return available

🚚 Delivery by 3 Oct 2023

COUPONS



1 Coupon applied

You saved additional ₹301

EDIT

GIFTING & PERSONALISATION



Yay! Gift Wrapping applied

Your order will be gift wrapped with your personalised message

REMOVE

EDIT MESSAGE

PRICE DETAILS (2 Items)

Total MRP ₹18,098

Discount on MRP -₹2,585

Coupon Discount ① -₹301

Gift Wrap Charges ₹25

Convenience Fee [Know More](#) ₹20

Total Amount ₹15,257

PLACE ORDER

8. Order of Operations

B	O	D	M	A	S
Bracket	Order	Divide	Multiply	Add	Subtract
()	$\sqrt{x}$ x^2	$\div$ or $\times$	$+$ or $-$		
Parentheses	Exponents	Multiply	Divide	Add	Subtract
P	E	M	D	A	S

$9 \div 3 \times 2 \div 6$

$8 - 5 + 7 - 1$

MYNTRA Cart

✓ 2/2 ITEMS SELECTED

REMOVE

MOVE TO WISHLIST



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GIFTING & PERSONALISATION



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REMOVE

EDIT MESSAGE

PRICE DETAILS (2 Items)

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Discount on MRP -₹2,585

Coupon Discount ① -₹301

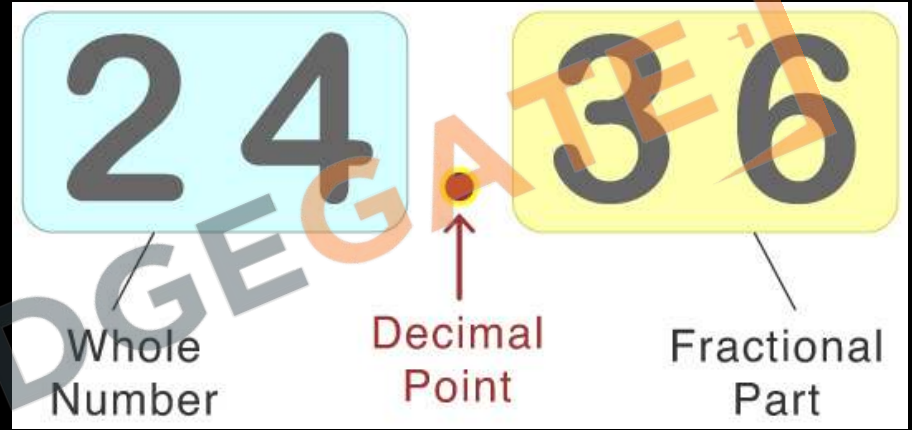
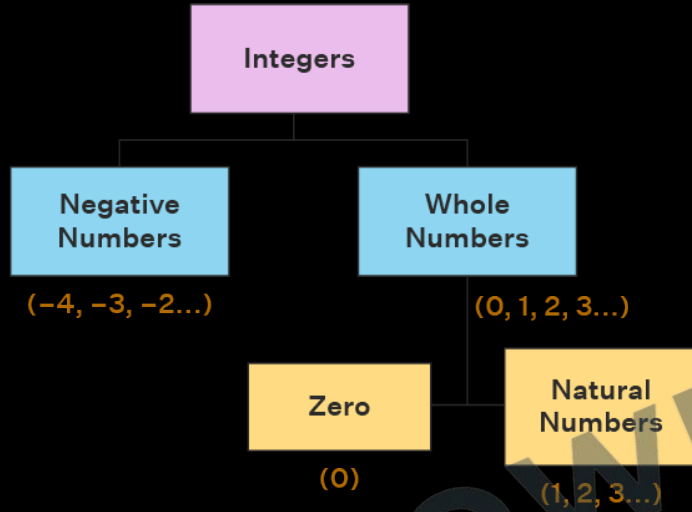
Gift Wrap Charges ₹25

Convenience Fee [Know More](#) ₹20

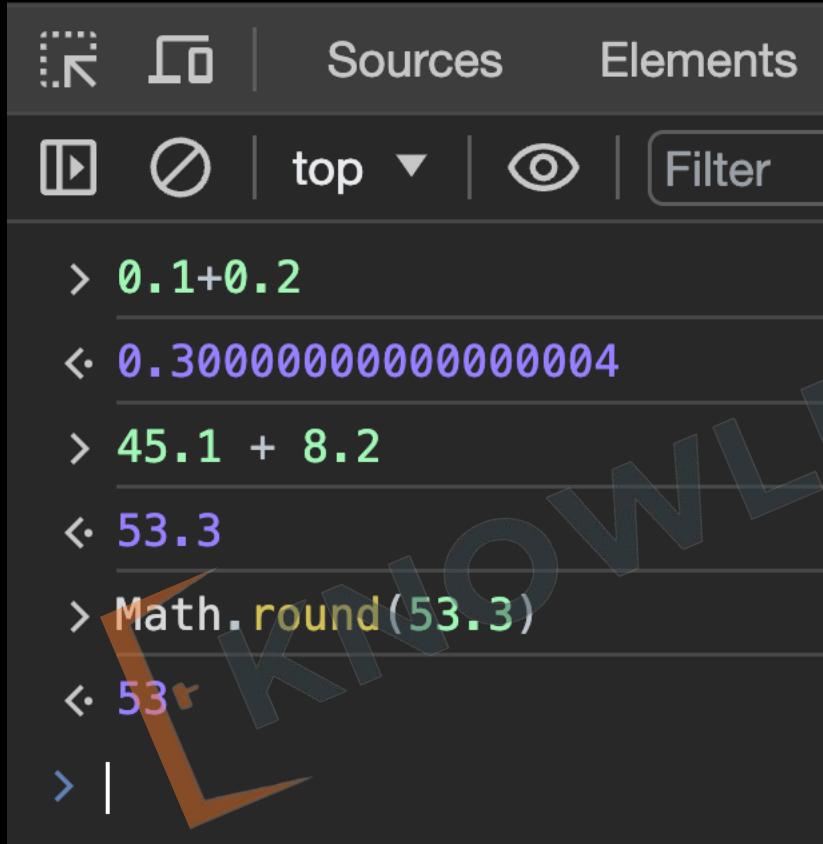
Total Amount ₹15,257

PLACE ORDER

9. Types of Numbers (Integers & Floats)



9. Types of Numbers (Float Problems)



```
> 0.1+0.2
< 0.30000000000000004
> 45.1 + 8.2
< 53.3
> Math.round(53.3)
< 53
> |
```

1. JavaScript has many problems with float operations.
2. We can use `Math.round()` to convert floats to integers.
3. Always do money calculation in Paisa instead of Rupees

10. Don't **learn** syntax.



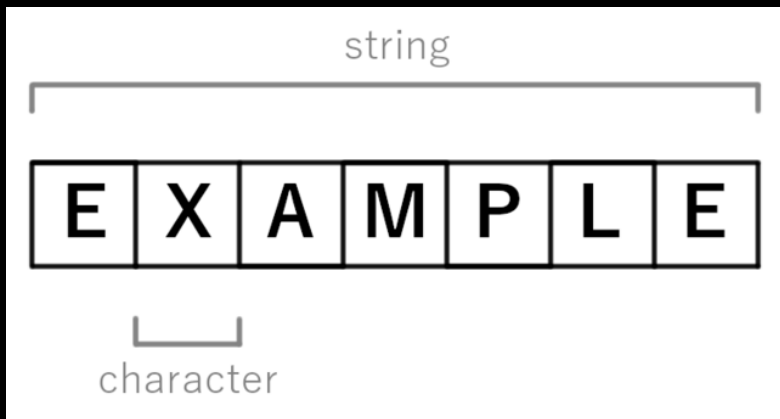
MDN web docs

moz://a



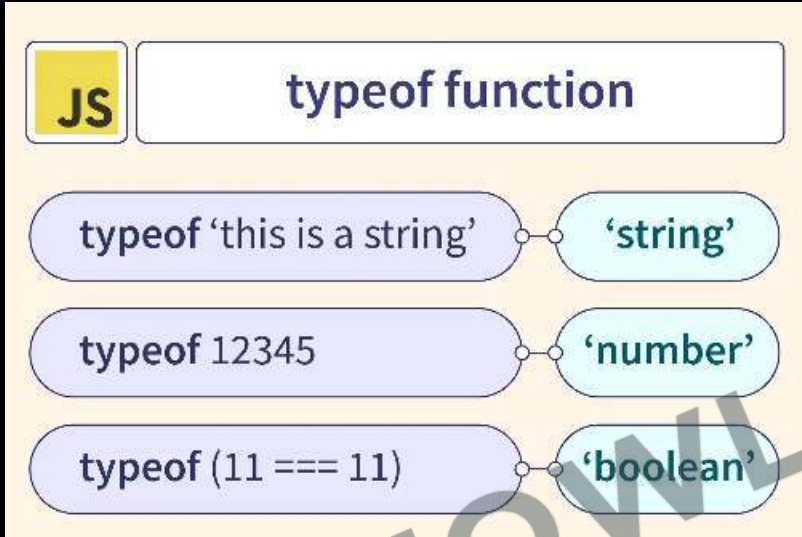
1. **Google**: Quick answers to coding problems.
2. **MDN**: In-depth guides and documentation.
<https://developer.mozilla.org/>
3. **ChatGPT**: Real-time assistance for coding queries.
4. **Focus**: Understand concepts, not just syntax.

11. Strings



1. Strings hold **textual data**, anything from a single character to paragraph.
2. Strings can be defined using **single quotes** `'`, **double quotes** `"`, or **backticks** ```. Backticks allow for template literals, which can include variables.
3. You can combine (**concatenate**) strings using the `+` operator. For example, `"Hello" + " World"` will produce `"Hello World"`.

12. Type Of Operator

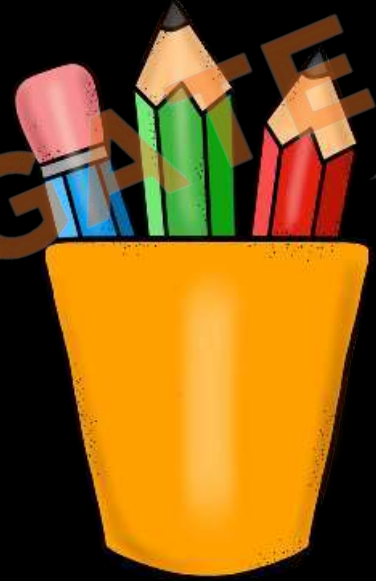


1. **Check Type:** Tells you the data type of a variable.
2. **Syntax:** Use it like `typeof` variable.
3. **Common Types:** Returns `"number,"` `"string,"` `"boolean,"` etc.

Numbers & Strings

Revision

7. Arithmetic Operators
8. Order of Operations
9. Types of Numbers
10. Don't learn syntax.
11. Strings
12. TypeOf Operator



Practice Exercise

Numbers & Strings

1. At a restaurant you ate: 1 Dal ₹ 100, 2 Roti ₹10 each, 1 Ice Cream ₹30, calculate and display the final bill amount.
2. Calculate 18% GST on iPhone15 ₹79,990 and 2 Air pods Pro ₹24990 each.
3. Create strings using all 3 methods.
4. Concatenate String with Strings, and String with numbers.
5. Create Order Summary String for our Myntra Cart.
6. Display order summary in a popup



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JavaScript With HTML & CSS

13. VS Code IDE
14. HTML Tags Introduction
15. CSS Introduction
16. Query Selector
17. Script Tag
18. Comments

13. What is IDE

1. IDE stands for Integrated Development Environment.
2. Software suite that consolidates basic tools required for software development.
3. Central hub for coding, finding problems, and testing.
4. Designed to improve developer efficiency.



13. Need of IDE

1. Streamlines development.
2. Increases productivity.
3. Simplifies complex tasks.
4. Offers a unified workspace.
5. IDE Features
 1. Code Autocomplete
 2. Syntax Highlighting
 3. Version Control
 4. Error Checking



The screenshot shows a code editor window titled 'MainActivity.kt'. The code is written in Kotlin and defines a composable function named 'MessageCard'. The function takes a 'Message' object as a parameter. Inside the function, there is a 'Row' composable that contains an 'Image' and a 'Column'. The 'Image' composable uses 'painterResource' to load the 'android_studio_logo' and sets its 'contentDescription' to 'Profile Picture'. The 'Column' composable has a white background and contains a 'Text' element for the author's name and another 'Text' element for the message body. The code is syntax-highlighted, with keywords in blue, strings in red, and comments in green. A large 'KNOWLEDGEGATE' watermark is visible across the image.

```
MainActivity.kt

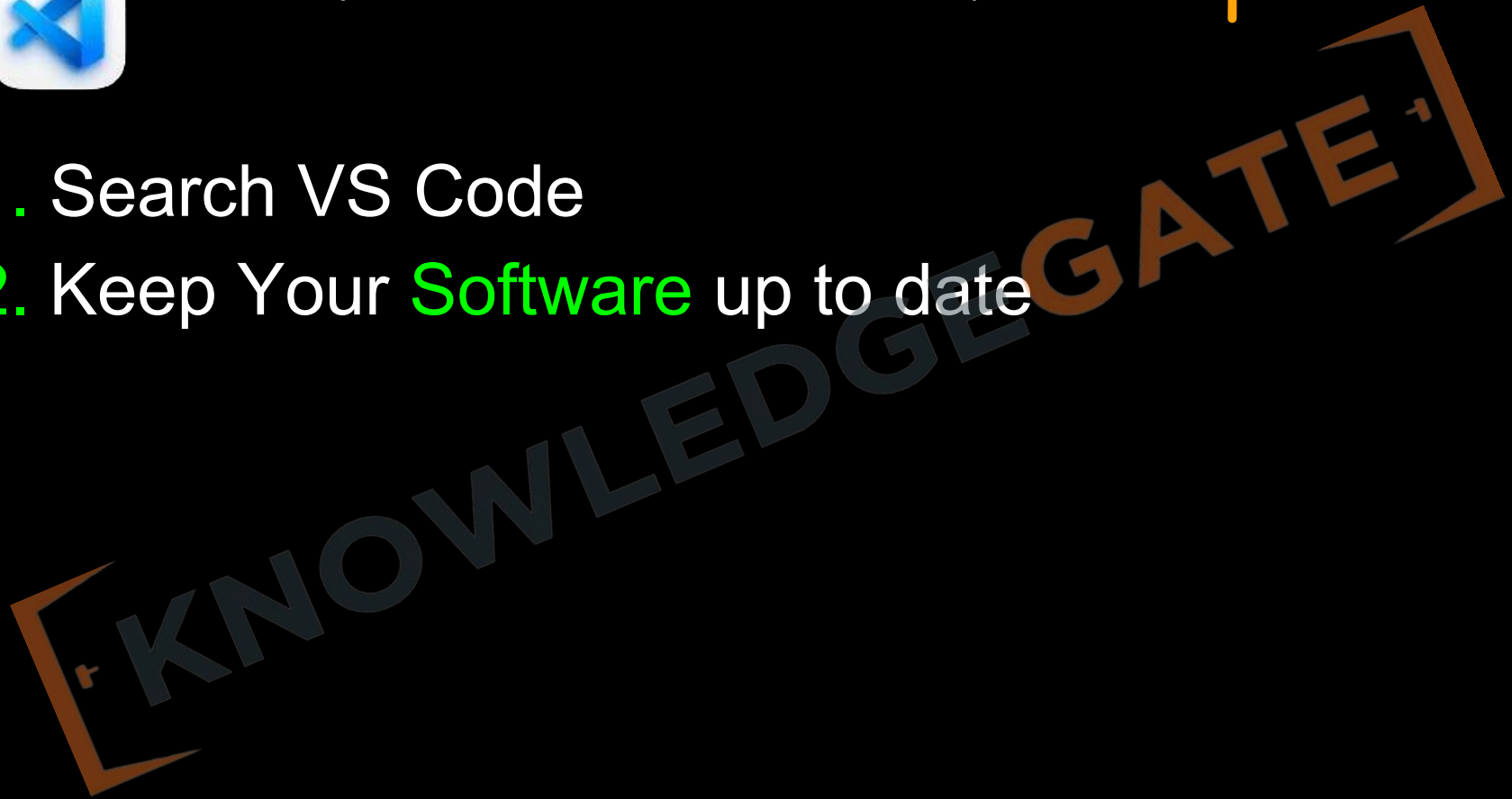
@Composable
fun MessageCard(msg: Message) {
    Row(modifier = Modifier.padding(all = 8.dp)) {
        Image(
            painter = painterResource(R.drawable.android_studio_logo),
            contentDescription = "Profile Picture",
            modifier = Modifier
                .size(45.dp)
        )

        Spacer(modifier = Modifier.width(8.dp))
        Column(modifier =
            .background(color = [Color.White])) {
            Text(text = msg.author, color = Color.Black)
            Spacer(modifier = Modifier.height(1.dp))
            Text(text = msg.body, color = Color.Black)
        }
    }
}
```

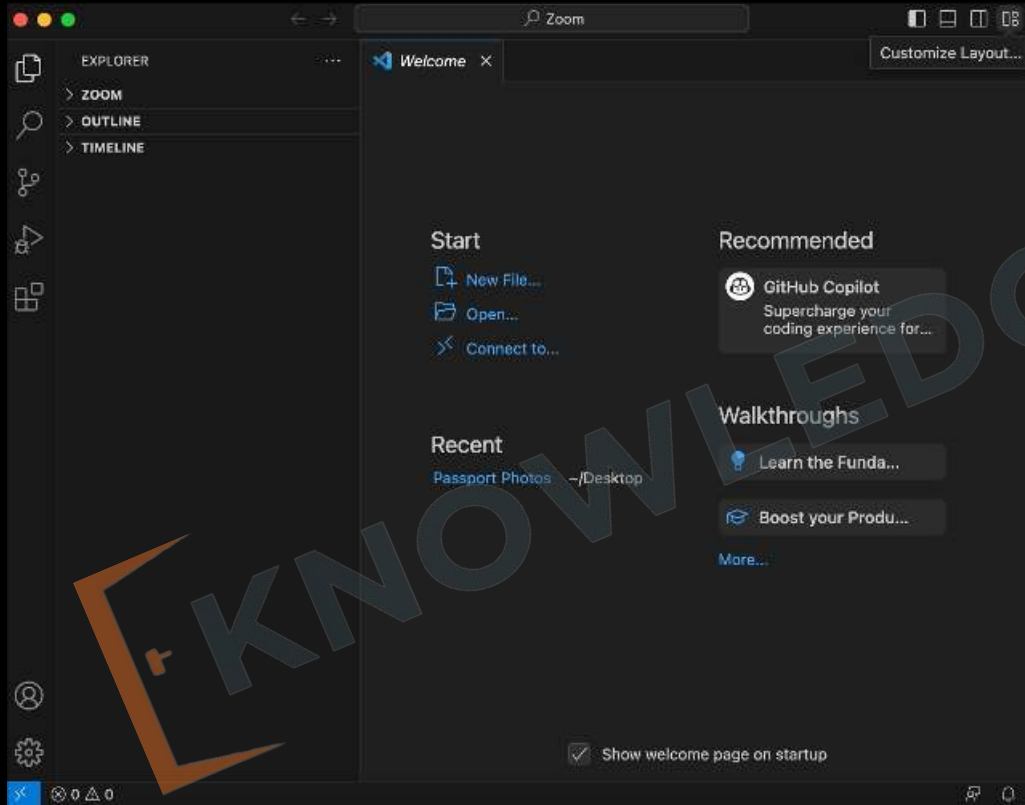


13. Installation & Setup

1. Search VS Code
2. Keep Your Software up to date

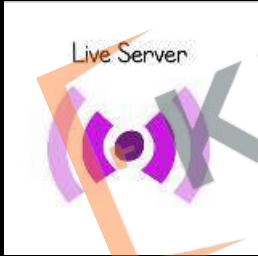


13. Opening project in VsCode



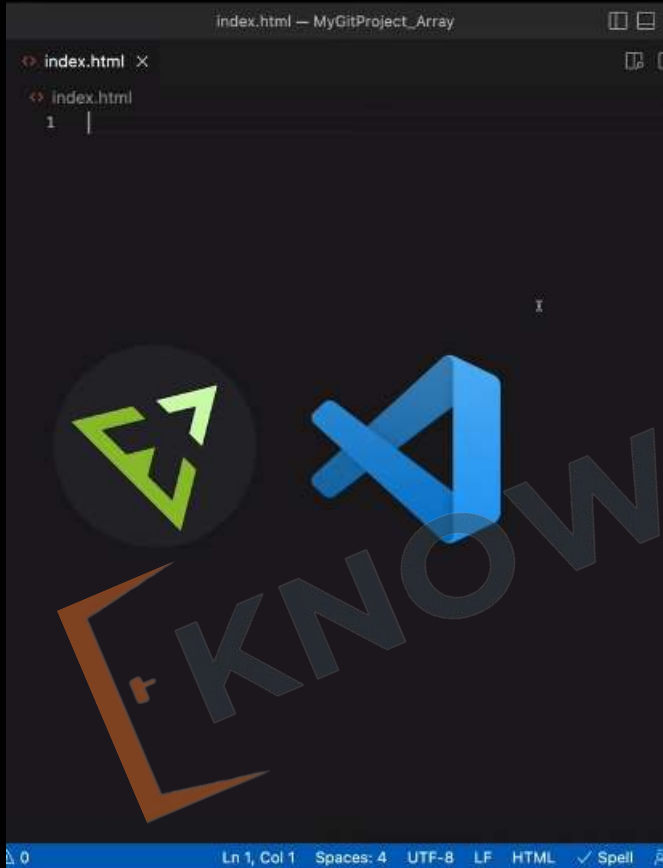
13. VsCode Extensions and Settings

1. Live Server
2. Prettier
3. Line Wrap
4. Tab Size from 4 to 2



13. Using Emmet ! to generate code

1. Type ! and wait for suggestions.

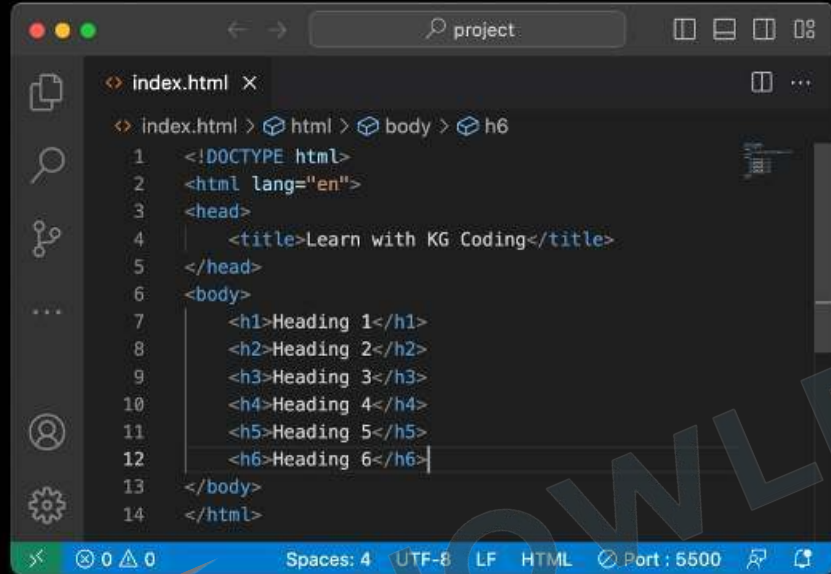


14 HTML Tags Introduction

1. **Elements** that are used to create a website are called HTML Tags.
2. Tags can contain **content** or **other HTML tags**.
3. Define elements like **text**, **images**, **links**



14 Heading Tag



```
<?index.html>
<?index.html> <?html> <?body> <?h6>
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <title>Learn with KG Coding</title>
5  </head>
6  <body>
7    <h1>Heading 1</h1>
8    <h2>Heading 2</h2>
9    <h3>Heading 3</h3>
10   <h4>Heading 4</h4>
11   <h5>Heading 5</h5>
12   <h6>Heading 6</h6>
13 </body>
14 </html>
```



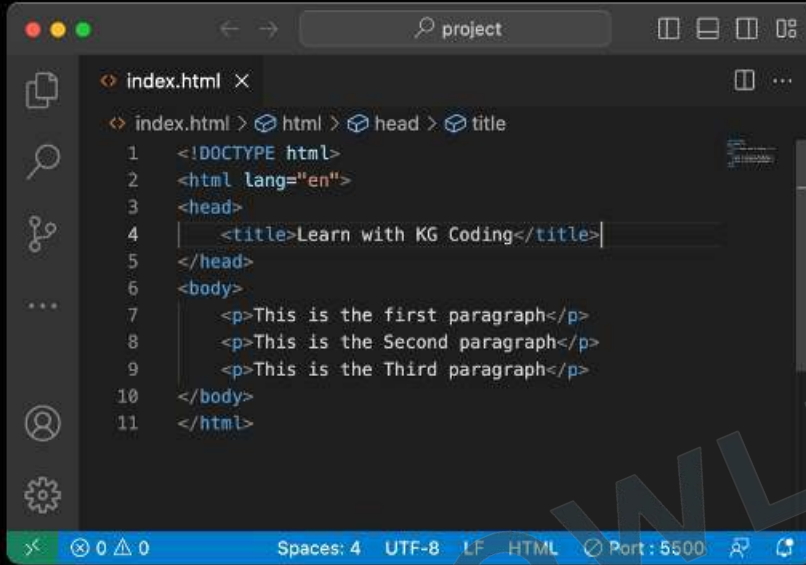
1. Defines **headings** in a document
2. Ranges from **<h1>** to **<h6>**
3. **<h1>** is most important, **<h6>** is least
4. Important for **SEO**
5. Helps in structuring content

14 Button tag

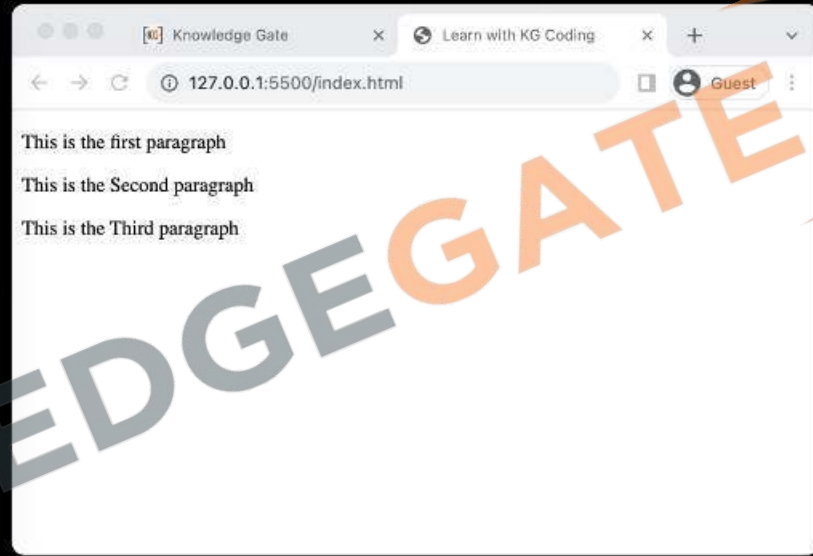
```
Button.html — testing
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Button</title>
  <style>
    button {
      margin: 10px;
    }
    .blue {
      background-color: blue;
      color: red;
    }
    .wow {
      background-color: #4CAF50;
      border: none;
      border-radius: 10px;
      color: white;
      padding: 8px 10px;
    }
  </style>
</head>
<body>
  <button>Click Me</button> <br>
  <button class="blue">Blue button</button> <br>
  <button class="wow">Sundar button</button>
</body>
</html>
```



14 Paragraph Tag

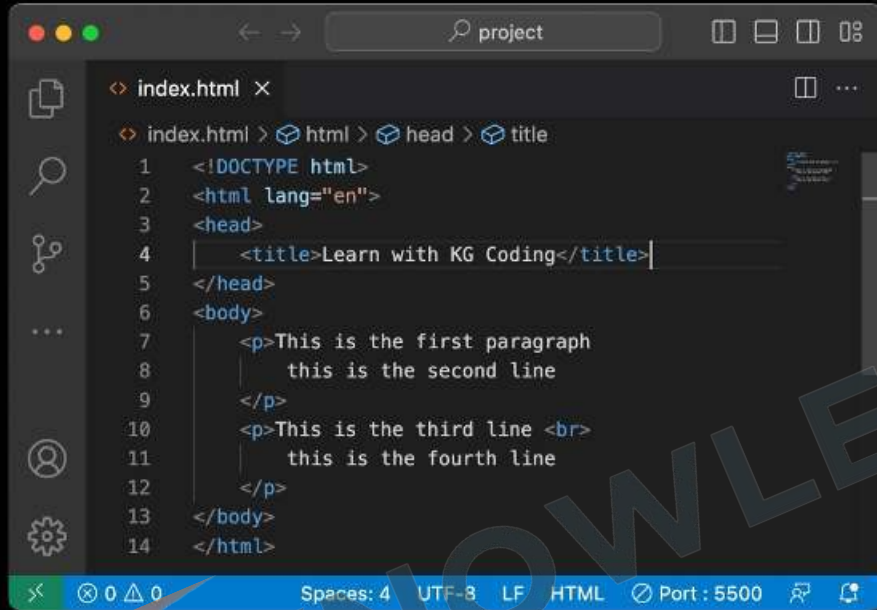


```
index.html X
index.html > html > head > title
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>Learn with KG Coding</title>
5 </head>
6 <body>
7   <p>This is the first paragraph</p>
8   <p>This is the Second paragraph</p>
9   <p>This is the Third paragraph</p>
10 </body>
11 </html>
```

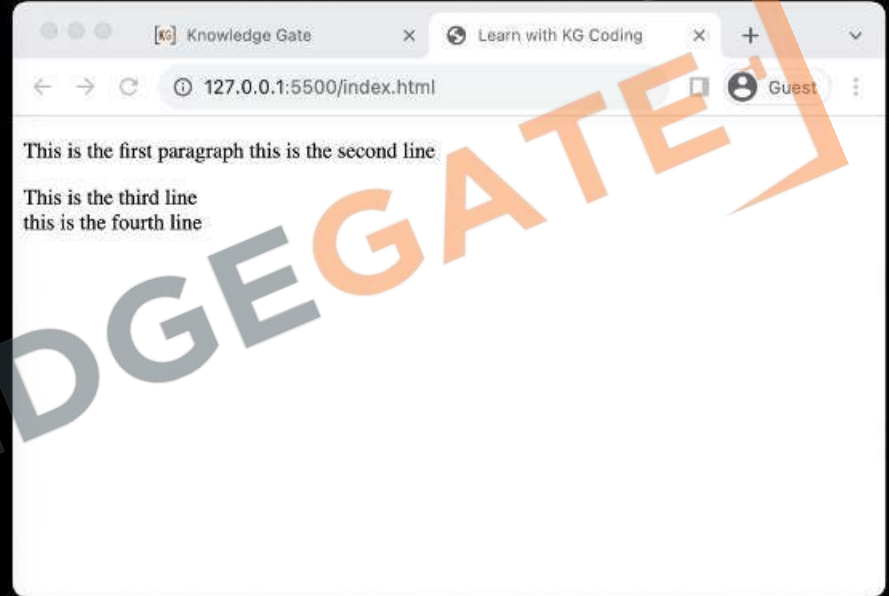


1. Used for defining paragraphs
2. Enclosed within `<p>` and `</p>` tags
3. Adds automatic spacing before and after
4. Text wraps to next line inside tag
5. Common in text-heavy content

14
 Tag



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>Learn with KG Coding</title>
5 </head>
6 <body>
7   <p>This is the first paragraph
8     this is the second line
9 </p>
10  <p>This is the third line <br>
11    this is the fourth line
12 </p>
13 </body>
14 </html>
```

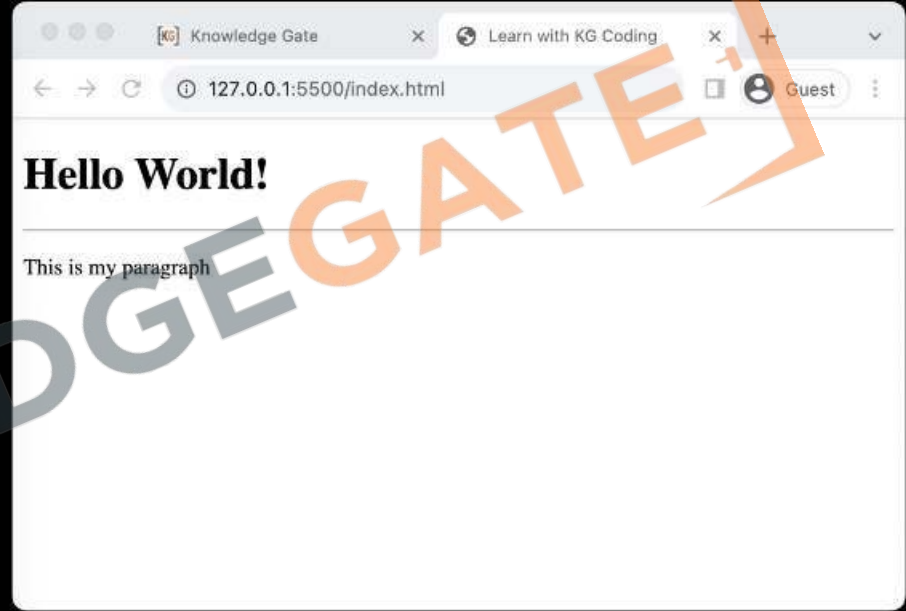


1. **
** adds a **line break** within text
2. **
** is empty, no closing tag needed
3. **
** and **
** are both valid

14 <HR> Tag

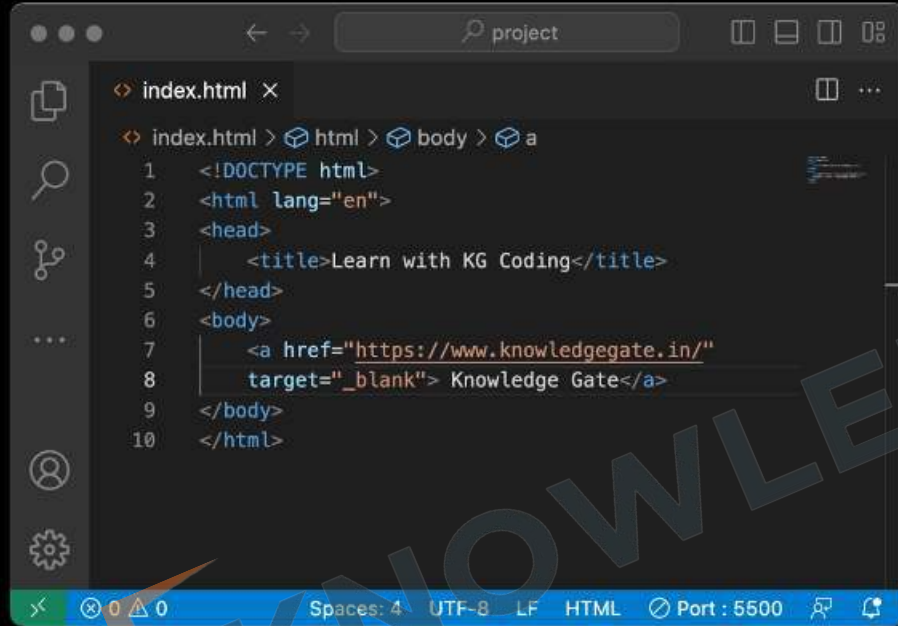


```
<code>index.html x
index.html > html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>Learn with KG Coding</title>
5 </head>
6 <body>
7   <h1>Hello World!</h1>
8   <hr>
9   <p>This is my paragraph</p>
10 </body>
11 </html>|</code>
```



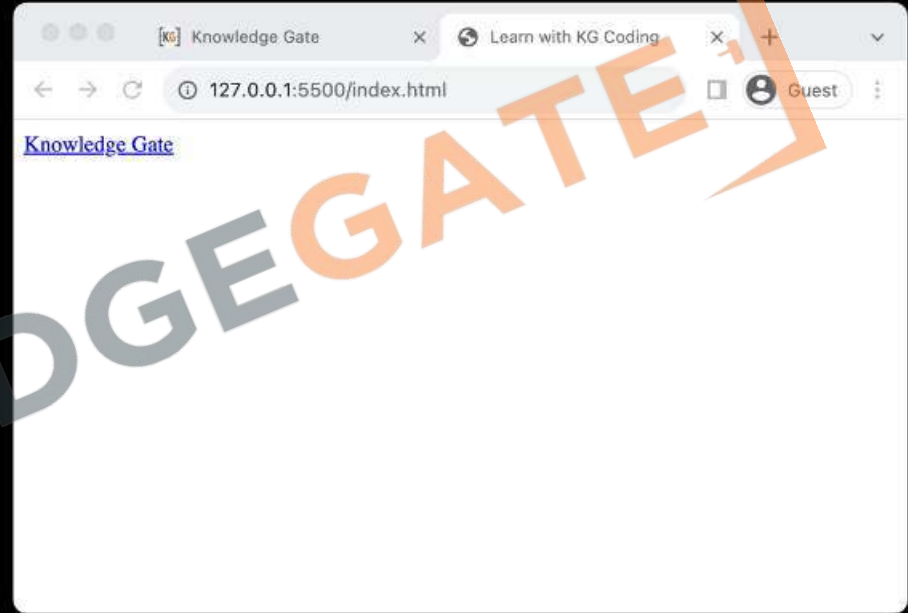
1. `<hr>` creates a horizontal rule or line
2. `<hr>` also empty, acts as a divider

14 Anchor Tag



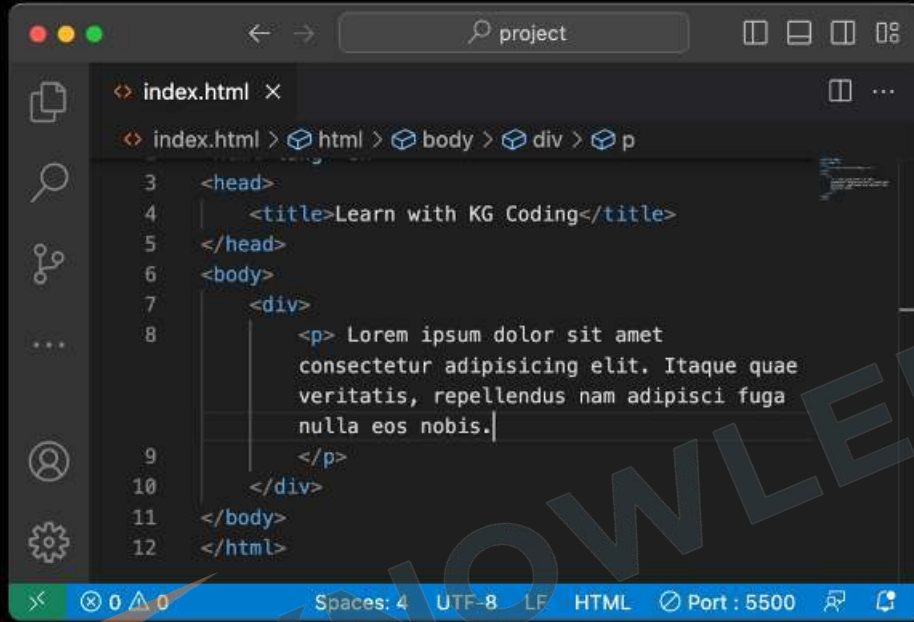
```
<?xml version="1.0" encoding="UTF-8" ?>
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Learn with KG Coding</title>
</head>
<body>
  <a href="https://www.knowledgegate.in/"
    target="_blank"> Knowledge Gate</a>
</body>
</html>
```

The screenshot shows a code editor with a dark theme. The file is named 'index.html'. The code is valid HTML5, including a DOCTYPE declaration, an HTML lang attribute, a head section with a title, and a body section containing an anchor tag. The anchor tag has a href attribute pointing to 'https://www.knowledgegate.in/' and a target attribute set to '_blank'. The editor's status bar at the bottom indicates 'Spaces: 4', 'UTF-8', 'LF', 'HTML', and 'Port : 5500'.

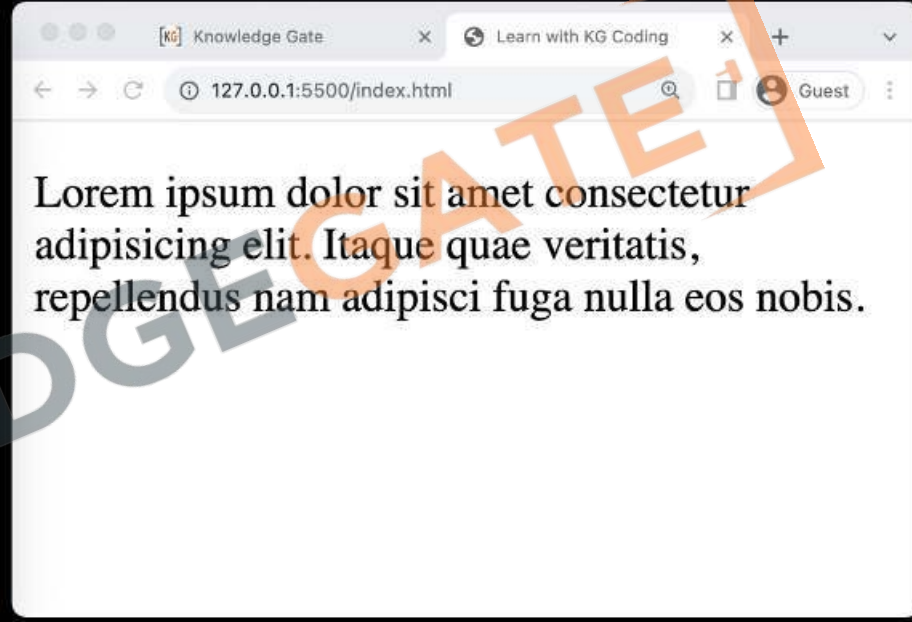


1. Used for creating **hyperlinks**
2. Requires **href** attribute for URL
3. Can link to external sites or internal pages
4. Supports **target** attribute to control link behavior

14 Div Tags



```
index.html x
index.html > html > body > div > p
3 <head>
4   <title>Learn with KG Coding</title>
5 </head>
6 <body>
7   <div>
8     <p> Lorem ipsum dolor sit amet
      consectetur adipiscing elit. Itaque quae
      veritatis, repellendus nam adipisci fuga
      nulla eos nobis.
9   </p>
10  </div>
11 </body>
12 </html>
```



1. **Purpose:** Acts as a container for other **HTML** elements.
2. **Non-Semantic:** Doesn't provide inherent meaning to enclosed content.
3. **Styling:** Commonly used for layout and styling via **CSS**.
4. **Flexibility:** Highly versatile and can be customized using classes or IDs.

14 Basic HTML Page

```
<!DOCTYPE html>
```

Defines the HTML Version

```
<html lang="en">
```

Parent of all HTML tags / Root element

```
<head>
```

Parent of meta data tags

```
<title>My First Webpage</title>
```

Title of the web page

```
</head>
```

```
<body>
```

Parent of content tags

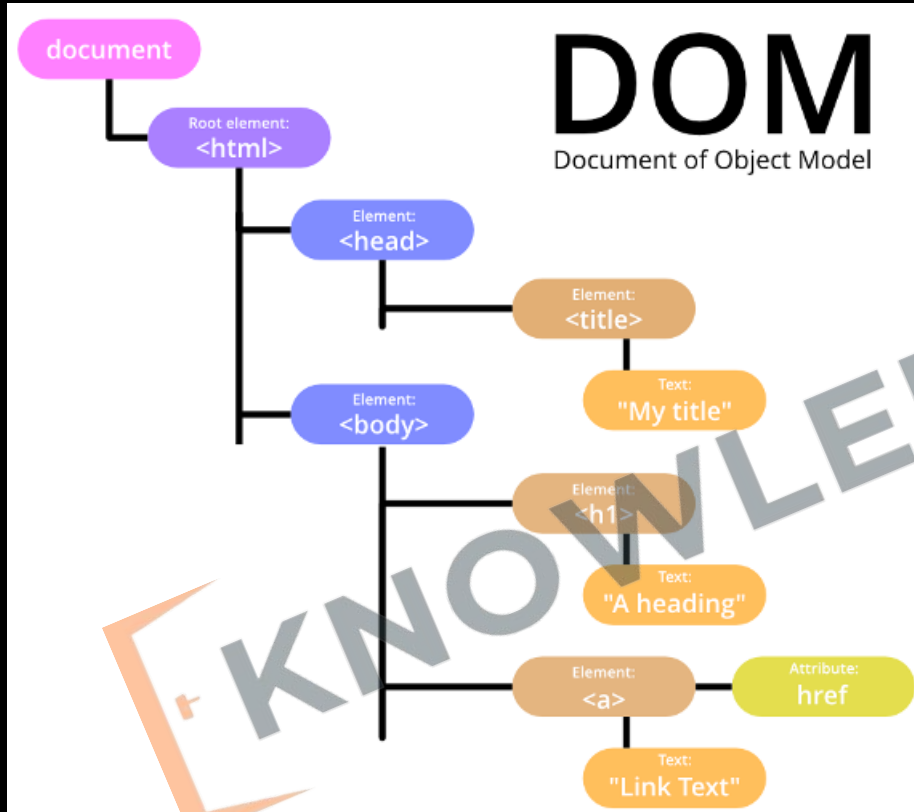
```
<h1>Hello World!</h1>
```

Heading tag

```
</body>
```

```
</html>
```

14 HTML DOM



- 1. Structure Understanding:** Helps in understanding the **hierarchical structure** of a webpage, crucial for applying targeted CSS styles.
- 2. Dynamic Styling:** Enables learning about dynamic styling, allowing for **real-time changes** and interactivity through CSS.

14 HTML (Attributes)

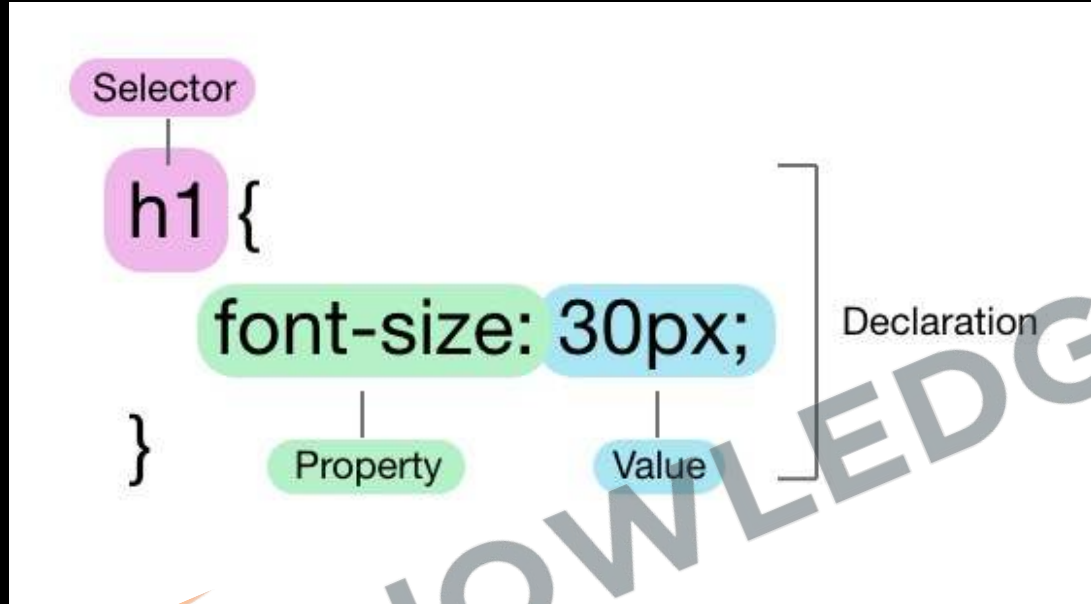
Html Attributes

Attribute

`<tag attribute="value">Text Content </tag>`

1. Provides **additional information** about elements
2. Placed **within opening tags**
3. Common examples: **href, src, alt**
4. Use **name=value** format
5. Can be **single** or **multiple** per element

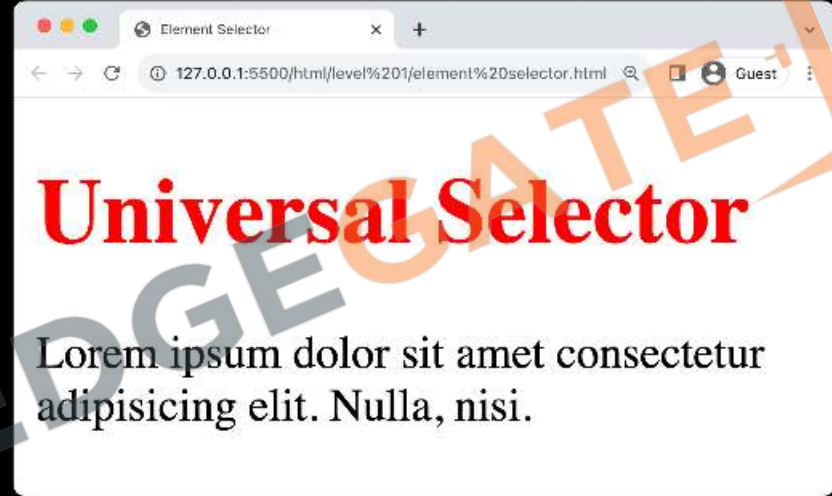
15 CSS (Basic Syntax)



- **Selector:** The HTML element that you want to style.
- **Property:** The attribute you want to change (like font, color, etc.).
- **Value:** The specific style you want to apply to the property (like red, bold, etc.).

15 CSS (Element selector)

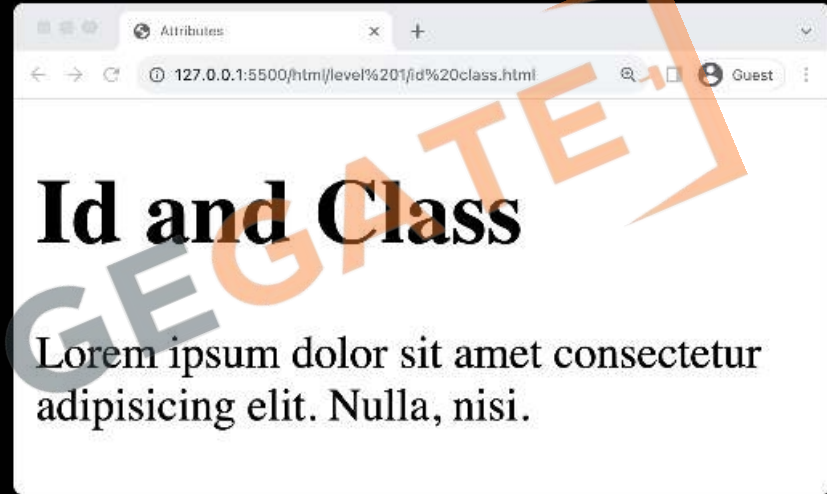
```
3 <head>
4   <title>Element Selector</title>
5   <style>
6     h1 {
7       color: red
8     }
9   </style>
10 </head>
11 <body>
12   <h1>Universal Selector</h1>
13   <p>Lorem ipsum dolor sit amet consectetur
14     adipiscing elit. Nulla, nisi.</p>
15 </body>
```



- **Targets Elements:** Selects HTML elements based on their **tag name**.
- **Syntax:** Simply use the **element's name**
- **Uniform Styling:** Helps in applying **consistent styles** to all instances.
- **Ease of Use:** Straightforward and **easy** to implement for basic styling.

15 CSS (id & class attribute)

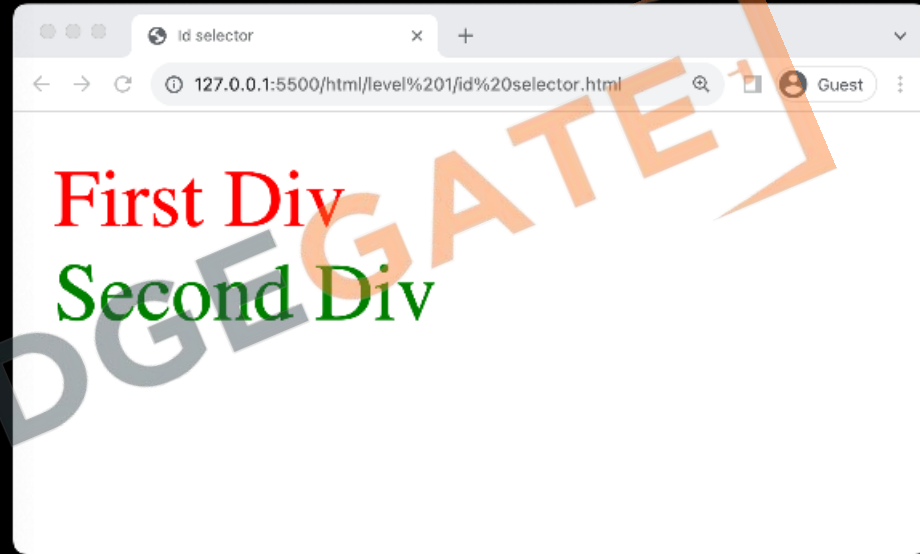
```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>Attributes</title>
5 </head>
6 <body>
7   <h1 id="top_heading">Id and Class</h1>
8   <p class="article">Lorem ipsum dolor sit amet
9     consectetur adipisicing elit. Nulla, nisi.</p>
10 </body>
</html>
```



- **ID Property:** Assigns a unique identifier to a single HTML element.
- **Class Property:** Allows grouping of multiple HTML elements to style them collectively.
- **Reusable Classes:** Class properties can be reused across different elements for consistent styling.
- **Specificity and Targeting:** Both properties assist in targeting specific elements or groups of elements for precise styling.

15 CSS (Id selector)

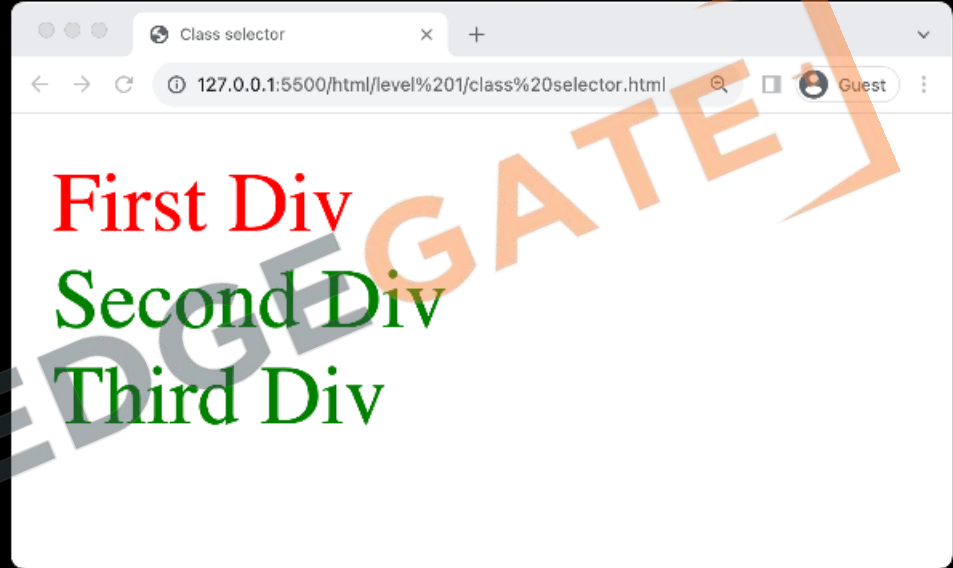
```
3 <head>
4   <title>Id selector</title>
5   <style>
6     #first { color: red; }
7     #second { color: green; }
8   </style>
9 </head>
10 <body>
11   <div id="first">First Div</div>
12   <div id="second">Second Div</div>
13 </body>
```



- **Unique Identifier:** Targets a specific element with a **unique ID** attribute.
- **Syntax:** Uses the **hash (#)** symbol
- **Single Use:** Each ID should be used **once per page** for uniqueness.
- **Specific Targeting:** Ideal for styling **individual, distinct** elements.

15 CSS (Class selector)

```
<head>
  <title>Class selector</title>
  <style>
    #first { color: red; }
    .second { color: green; }
  </style>
</head>
<body>
  <div id="first">First Div</div>
  <div class="second">Second Div</div>
  <div class="second">Third Div</div>
</body>
```

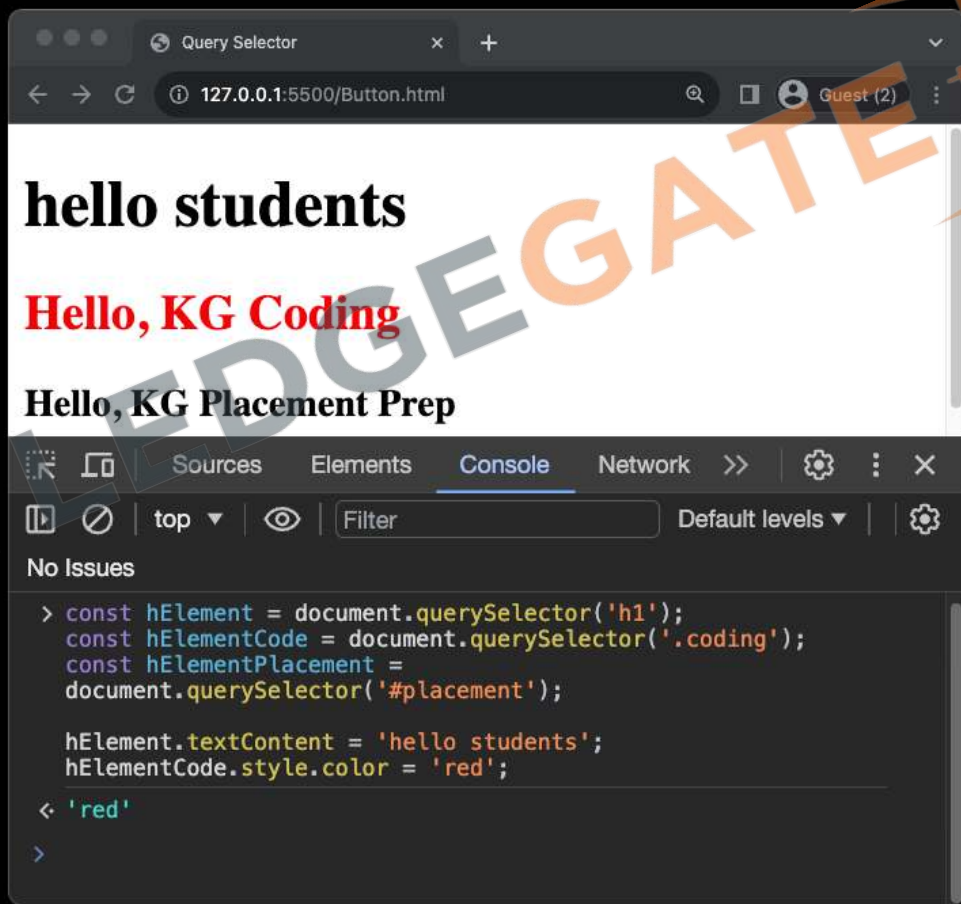
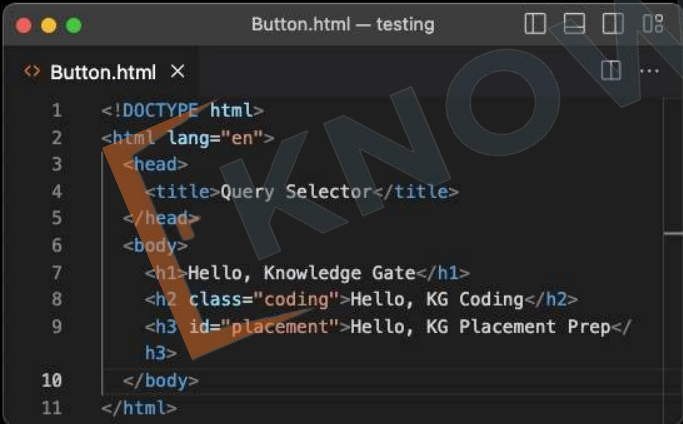
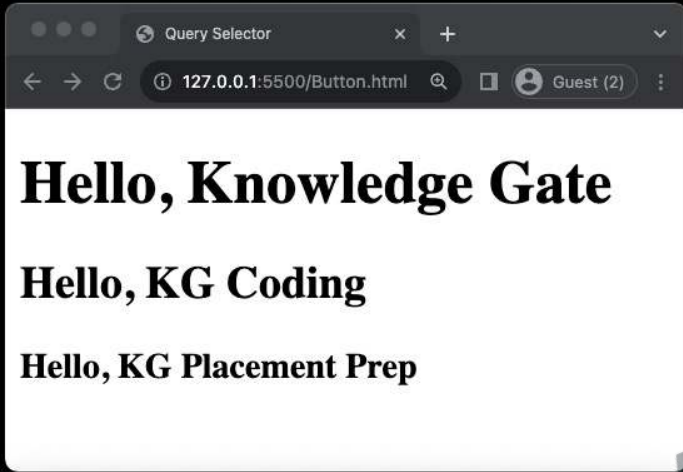


- **Group Styling:** Allows styling of multiple elements grouped under a class.
- **Syntax:** Utilizes the dot (.) symbol.
- **Reusable:** Can be used on multiple elements for consistent styling.
- **Versatility:** Ideal for applying styles to a category of elements.

16 Query Selector

1. **getElementById**: Finds one element by its ID.
2. **getElementsByClassName**: Finds elements by their class, returns a list.
3. **querySelector**: Finds the first element that matches a CSS selector.
4. **Purpose**: To interact with or modify webpage elements.

16 Query Selector

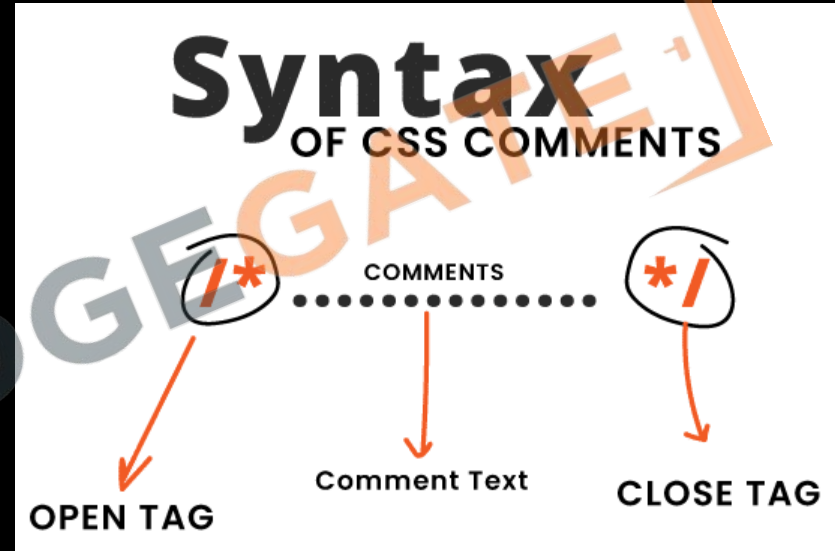


17 Script Tag

1. **Embed Code:** Incorporates JavaScript into an HTML file, either directly or via external files.
2. **Placement:** Commonly placed in the `<head>` or just before the closing `</body>` tag to control when the script runs.
3. **External Files:** Use `src` attribute to link external JavaScript files, like `<script src="script.js"></script>`.
4. **Console Methods:** `log`, `warn`, `error`, `clear`

18 JavaScript & CSS Comments

1. Used to add **notes in source code** in JavaScript or CSS.
2. **Not displayed** on the web page
3. Syntax: `/* comment here */`
4. Helpful for **code organization**
5. Can be **multi-line** or **single-line**



18 HTML Comments

1. Used to add **notes** in HTML code
2. **Not displayed** on the web page
3. Syntax: **<!-- Comment here -->**
4. Helpful for **code organization**
5. Can be **multi-line** or **single-line**

Writing comments in HTML

Single-line
Comment

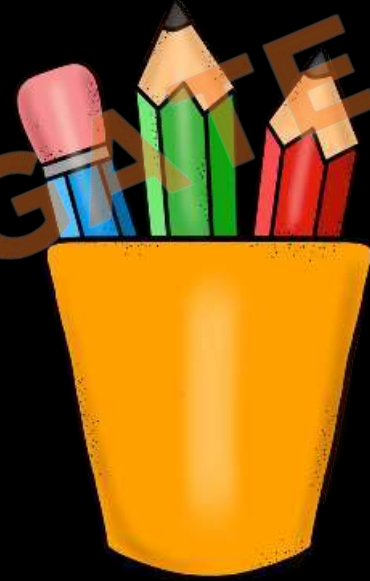
```
1 <!--This is a single line  
comment in HTML. You cannot  
see it on a webpage. Click  
on view-source to see a  
message I left just for you.  
-->
```

Multi-line
Comment

```
1 <!-- This is a multi-line  
comment in HTML.  
2 You cannot see it on a  
webpage.  
3 If you view-source on the  
browser you can see the  
comment there.-->
```

JavaScript With HTML & CSS Revision

13. VS Code IDE
14. HTML Tags Introduction
15. CSS Introduction
16. Query Selector
17. Script Tag
18. Comments



Practice Exercise

JavaScript With HTML & CSS

1. Create a button with text click
2. Create 2 buttons with class and id
3. Create a paragraph
4. Add colors to two buttons
5. Add proper html structure
6. Change page title
7. Try to copy the given design on the bottom



8. Add script element to page to show welcome alert
9. Add onclick alert on Add to Bag and ❤️ Wishlist buttons



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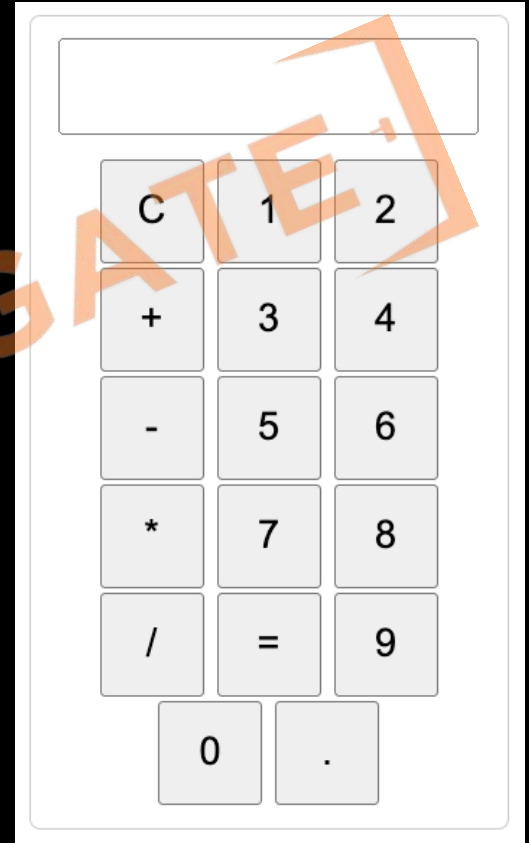
[KG Placement Prep](#)



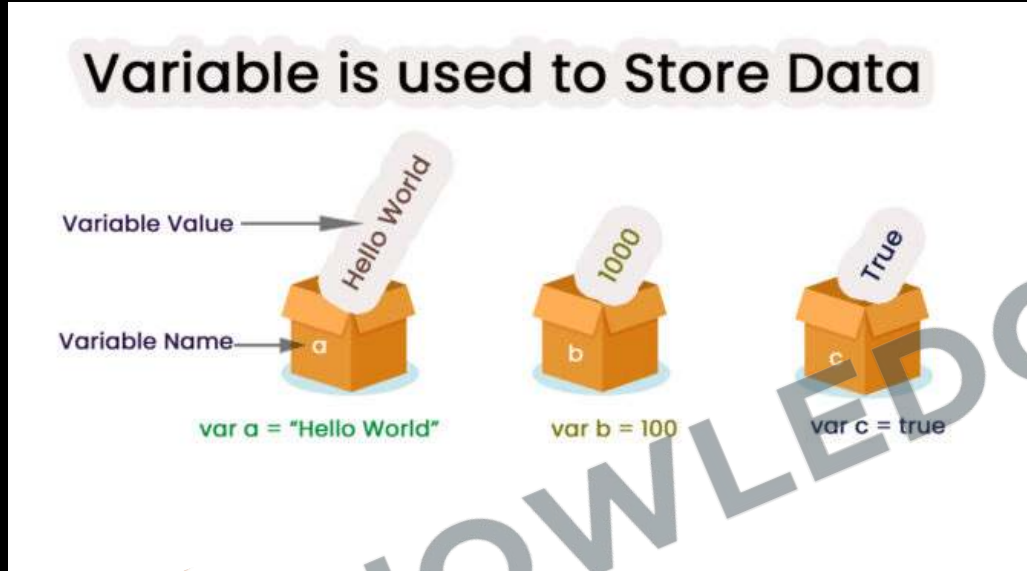
[Sanchit Socket](#)

Variables

- 19. What are Variables
- 20. Syntax Rules
- 21. Updating Values
- 22. Mynta Bag Exercise
- 23. Naming Conventions
- 24. Ways to create Variables



19. What are Variables?



Variables are like containers used for storing data values.

20. Syntax Rules

```
1 // Defining a number variable
2 let noOfStudents = 5;
3 // Defining a String variable
4 let welcomeMessage = "Hello Beta"
```

1. Can't use **keywords** or reserved words
2. Can't start with a **number**
3. No special characters other than **\$** and **_**
4. **=** is for **assignment**
5. **;** means end of **instruction**

21. Updating Values

```
let noOfStudents = 5;  
noOfStudents = noOfStudents + 1;
```

```
let money = 1;  
money += 5; // money = 6  
money -= 2; // money = 4  
money *= 3; // money = 12  
money /= 4; // money = 3  
money++;    // money = 4
```

1. Do not need to use **let** again.
2. Syntax: **variable = variable + 1**
3. Assignment Operator is used **=**
4. Short Hand Assignment Operators:
+=, -=, *=, /=, ++

22. Myntra Bag Exercise

Add to Bag

MOVE TO WISHLIST

Add 1+1 sale item

Your Bag has 2 items

23. Naming Conventions

camelCase

- Start with a lowercase letter. Capitalize the first letter of each subsequent word.
- Example: `myVariableName`

snake_case

- Start with an lowercase letter. Separate words with **underscore**
- Example: `my_variable_name`

Kebab-case

- All lowercase letters. Separate words with **hyphens**. Used for HTML and CSS.
- Example: `my-variable-name`

Keep a Good and Short Name

- Choose names that are descriptive but not too long. It should make it easy to understand the variable's purpose.
- Example: `age`, `firstName`, `isMarried`



24. Ways to Create Variables

var

var apple = 



a thing in a box
named "apple"

apple = 



you can swap
item later

let

let apple = 



a thing in a box
named "apple" w/
protection shield

apple = 



OK!

you can swap item
only if you ask
inside of the shield

~~apple = ~~ NG

const

const apple = 



a thing in
LOCKED cage
named "apple"



~~apple = ~~ NG

you can't
swap item
later.



apple.multiply(3)
OK!

... but you can ask
the item to change itself
(if the item has method
to do that)

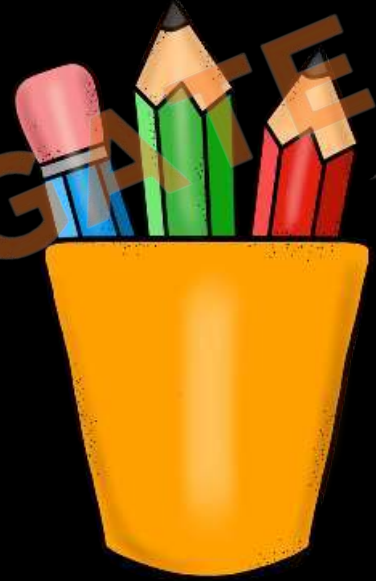
const

let

var

Variables Revision

19. What are Variables
20. Syntax Rules
21. Updating Values
22. Mynta Bag Exercise
23. Naming Conventions
24. Ways to create Variables



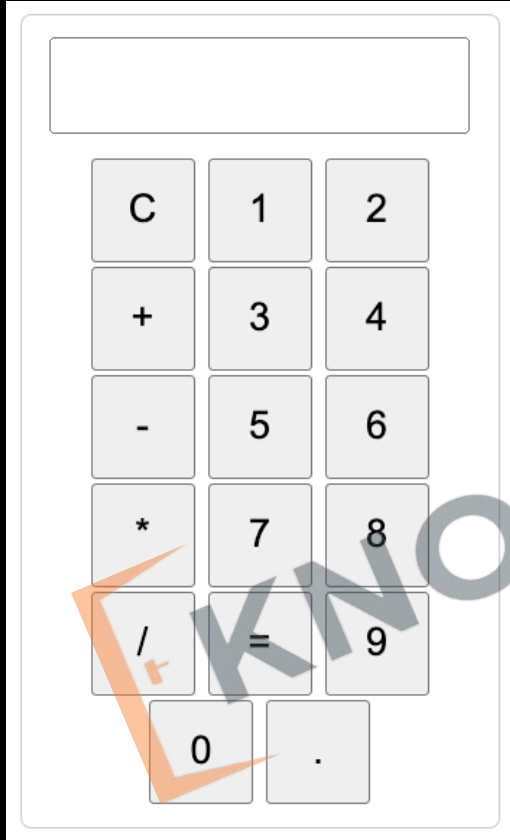
Practice Exercise

Variables

1. Save your name in a **variable** inside script tag
2. Display name from the **variable** on the page
3. Calculate the cost of **Myntra Bag** and keep it in a variable
4. Show it to **console**
5. Keep **GST** percentage as **constant**
6. Use **eval** method from math to convert string calculation into result



Project One: Calculator



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if-else & Boolean

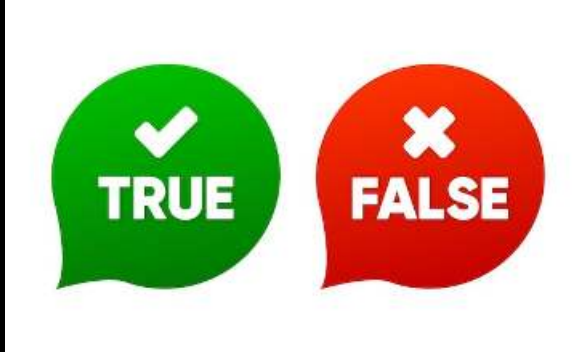
- 25. What are Booleans
- 26. Comparison Operators
- 27. if-else
- 28. Logical Operator
- 29. Scope
- 30. Truthy and Falsy Values
- 31. If alternates

Create



Project

25. What are Booleans?



1. **Data Type:** **Booleans** are a basic data type in JavaScript.
2. **Two Values:** Can only be **true** or **false**.
3. **'true'** is a String not a Boolean

26. Comparison Operators

<, >, <=, >=, ==, !=

Equality

- == Checks value equality.
- === Checks value and type equality.

Inequality

- != Checks value inequality.
- !== Checks value and type inequality.

Relational

- > Greater than.
- < Less than.
- >= Greater than or equal to.
- <= Less than or equal to.

Order of comparison operators is less than arithmetic operators

27. if-else

1. **Syntax:** Uses `if () {}` to check a condition.
2. **What is if:** Executes block if condition is **true**, skips if **false**.
3. **What is else:** Executes a block when the if condition is **false**.
4. **Curly Braces** can be omitted for single statements, but not recommended.
5. **If-else Ladder:** Multiple if and else if blocks; **only one** executes.
6. **Use Variables:** Can store **conditions** in variables for use in if statements.

```
if thirsty {  
      
}  
else {  
      
}
```

Project Cricket Game

Bat Ball Stump Game



Defeats



Bat

Ball

Stump



Defeats



Defeats



28. Logical Operators



AND

Or

NOT

1. Types: $\&\&$ (AND), $\|\|$ (OR), $!$ (NOT)
2. AND ($\&\&$): All conditions **must be true** for the result to be true.
3. OR ($\|\|$): Only **one condition** must be true for the result to be true.
4. NOT ($!$): **Inverts** the Boolean value of a condition.
5. Lower Priority than **Math** and **Comparison** operators

Project Cricket Game



Defeats



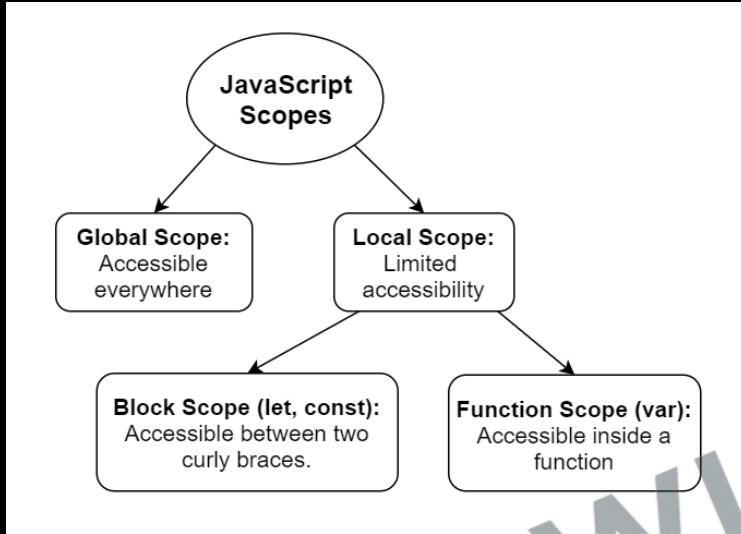
Defeats



Defeats



29. Scope



1. Any **variable created** inside `{ }` will remain inside `{ }`
2. **Variable** can be **redefined** inside `{ }`
3. **Var** does **not** follow **scope**
4. **Global Scope**: Accessible **everywhere** in the code.
5. **Block Scope**: **Limited** to a block, mainly with **let** and **const**.
6. **Declare variables** in the **narrowest** scope possible.

Project Cricket Game



Defeats



Defeats



Defeats



30. Truthy and Falsy Values

1. Falsy Values: 0, null, undefined, false, NaN, "" (empty string)
2. Truthy Values: All values not listed as falsy.
3. Used in conditional statements like if.
4. Non-boolean values are auto-converted in logical operations.
5. Be explicit in comparisons to avoid unexpected behaviour.



31. If alternates

1. Ternary Operator: `condition ? trueValue : falseValue`

Quick one-line if-else.

2. Guard Operator: `value || defaultValue`

Use when a `fallback` value is needed.

3. Default Operator: `value ?? fallbackValue`

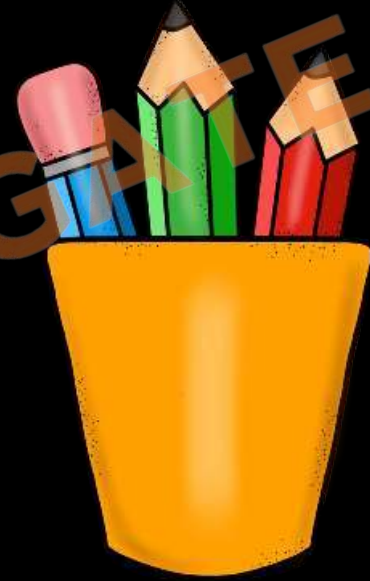
Use when you want to consider only null and undefined as falsy.

4. **Simplifies** conditional logic.

5. **Use** wisely to maintain readability.

if-else & Boolean Revision

- 25. What are Booleans
- 26. Comparison Operators
- 27. if-else
- 28. Logical Operator
- 29. Scope
- 30. Truthy and Falsy Values
- 31. If alternates



Practice Exercise

if-else & Boolean

1. Give discount based on age, gender for metro ticket
 - Females get 50% off
 - Kids under 5 years of age are free
 - Kids up to 8 years have half ticket
 - People Over 65 years of age only pay 30% of the ticket
 - In case of multiple discounts max discount will apply.



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Functions

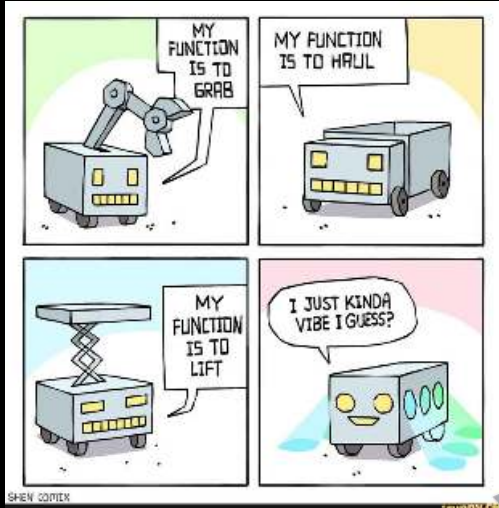
- 32. What are Functions
- 33. Function Syntax
- 34. Return statement
- 35. Function Parameters

Improve



Project

32. What are Functions?

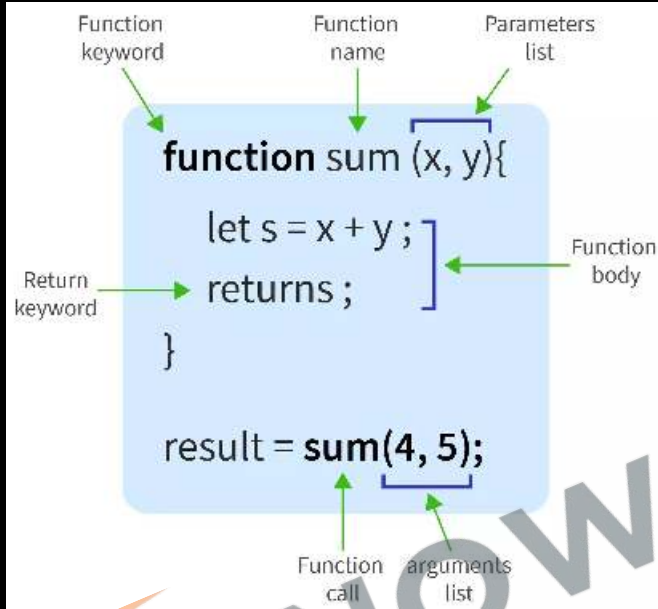


```
function greet(name) {  
    // code  
}  
  
greet(name);  
// code
```

function call

1. **Definition:** Blocks of reusable code.
2. **DRY Principle:** "Don't Repeat Yourself" it Encourages code reusability.
3. **Usage:** Organizes code and performs specific tasks.
4. **Naming Rules:** Same as variable names: camelCase
5. **Example:** "Beta Gas band kar de"

33. Functions Syntax



1. Use **function** keyword to declare.
2. Follows same rules as **variable names**.
3. Use **()** to contain parameters.
4. Invoke by using the **function name** followed by **()**.
5. **Fundamental** for **code organization** and **reusability**.

Project Cricket Game



Defeats



Create Method for Random Number &
Computer Choice



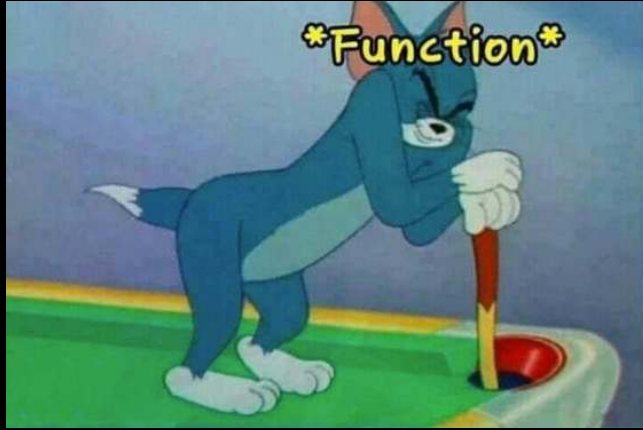
Defeats



Defeats



34. Return statement



1. Sends a value back from a function.
2. Example: "Ek glass paani laao"
3. What Can Be Returned: Value, variable, calculation, etc.
4. Return ends the function immediately.
5. Function calls make code jump around.
6. Prefer returning values over using global variables.

Project Cricket Game



Defeats



User return instead of Global Variable



Defeats



Defeats



35. Parameters



Parameter



Function



Return

1. **Input** values that a **function** takes.
2. Parameters put value into function, while return gets value out.
3. Example: "Ek packet dahi laao"
4. **Naming Convention**: Same as **variable** names.
5. **Parameter** vs **Argument**
6. Examples: **alert**, **Math.round**, **console.log** are functions we have already used
7. **Multiple Parameters**: Functions can take **more than one**.
8. **Default Value**: Can set a **default** value for a parameter.

Project Cricket Game



Defeats



Create method for comparing user choice & Showing Result alert



Defeats

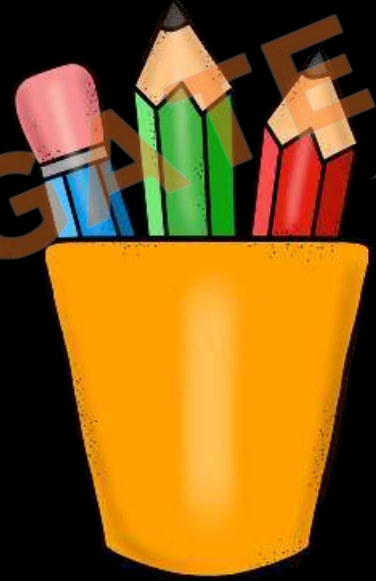


Defeats



Functions Revision

- 32. What are Functions
- 33. Function Syntax
- 34. Return statement
- 35. Function Parameters



Practice Exercise

Functions

1. Create a method to check if a number is odd or even.
2. Create a method to return larger of the two numbers.
3. Create Method to convert Celsius to Fahrenheit
$$F = (9/5) * C + 32$$



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Objects

- 36. What is an Object
- 37. Objects Syntax
- 38. Accessing Objects
- 39. Inside Objects
- 40. Autoboxing
- 41. Object References
- 42. Object Shortcuts

Maintain score of



Project

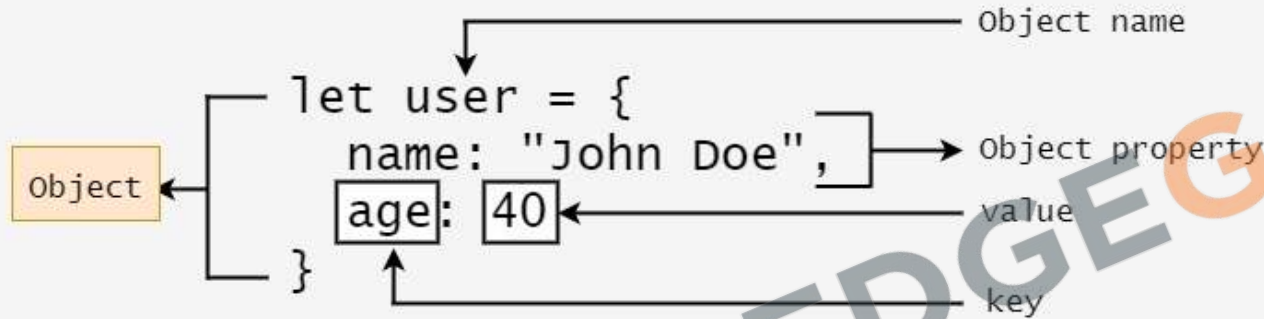
36. What is an Object?



```
let product = {  
  company: 'Mango',  
  item_name: 'Cotton striped t-shirt',  
  price: 861  
};
```

1. **Groups** multiple **values** together in **key-value** pairs.
2. **How to Define:** Use **{}** to enclose **properties**.
3. **Example:** product {**name**, **price**}
4. **Dot Notation:** Use **.** **operator** to access values.
5. **Key Benefit:** Organizes **related data** under a **single name**.

37. Object Syntax?



1. **Basic Structure**: Uses `{ }` to enclose **data**.
2. **Rules**: **Property** and **value** separated by a **colon(:)**
3. **Comma**: Separates different **property-value** pairs.
4. **Example**: `{ name: "Laptop", price: 1000 }`

38. Accessing Objects



1. **Dot Notation:** Access **properties** using **.** **Operator** like **product.price**
2. **Bracket Notation:** Useful for **properties** with **special characters** **product["nick-name"]**. Variables can be used to access properties
3. **typeof** returns **object**.
4. **Values** can be **added** or **removed** to an object
5. **Delete** Values using **delete**

Project Cricket Game



Defeats



Create object for maintaining Score



Defeats



Defeats



39. Inside Object

```
let product = {  
  company: 'Mango',  
  itemName: 'Cotton striped t-shirt',  
  price: 861,  
  rating: {  
    stars: 4.5,  
    noOfReviews: 87  
  },  
  displayPrice: function() {  
    return `$$${this.price.toFixed(2)}`;  
  }  
};
```



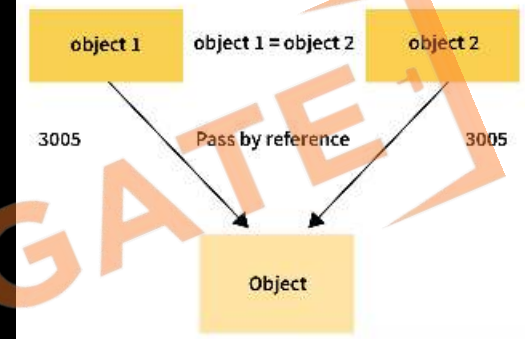
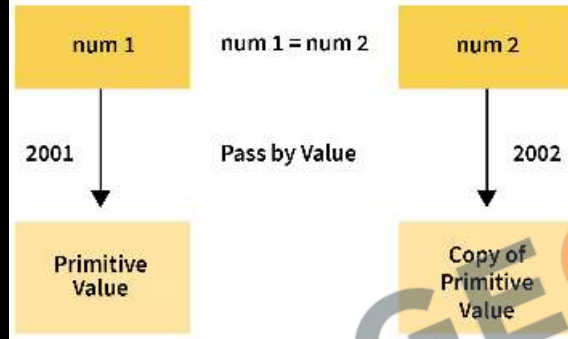
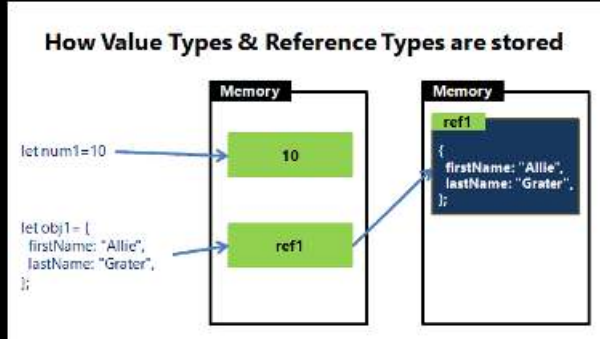
1. **Objects** can contain **Primitives** like **numbers** and **strings**.
2. **Objects** can contain **other objects** and are called Nested Objects.
3. **Functions** can be object properties.
4. **Functions** inside an object are called **methods**.
5. **Null Value**: **Intentionally** leaving a property **empty**.

40. Autoboxing



1. Automatic conversion of primitives to objects.
2. Allows properties and methods to be used on primitives.
3. Example: Strings have properties and methods like length, toUpperCase, etc.

41. Object References



1. **Objects** work based on **references**, not actual data.
2. **Copying** an object copies the reference, not the actual object.
3. **When comparing** with **==**, you're comparing references, not content.
4. **Changes** to one reference **affects all copies**.

42. Object Shortcuts

```
let product = {  
  company: 'Mango',  
  itemName: 'Cotton striped t-shirt',  
  price: 861  
};  
// Destructuring  
let company = product.company  
// is same as  
let { company } = product;
```

```
// Property shorthand  
let price = 861;  
let product = {  
  company: 'Mango',  
  itemName: 'Cotton striped t-shirt',  
  price: price  
};  
// is same as  
let product1 = {  
  company: 'Mango',  
  itemName: 'Cotton striped t-shirt',  
  price  
};
```

```
// Method shorthand  
let product = {  
  company: 'Mango',  
  itemName: 'Cotton striped t-shirt',  
  displayPrice: function() {  
    return `$$${this.price.toFixed(2)}`;  
  }  
};  
// is same as  
let product1 = {  
  company: 'Mango',  
  itemName: 'Cotton striped t-shirt',  
  displayPrice() {  
    return `$$${this.price.toFixed(2)}`;  
  }  
};
```

1. **De-structuring:** Extract properties from objects easily.
2. We can extract more than one property at once.
3. **Shorthand Property:** {message: message} simplifies to just message.
4. **Shorthand Method:** Define methods directly inside the object without the function keyword.

Project Cricket Game



Defeats



Move method for showing result into score
Object.



Defeats

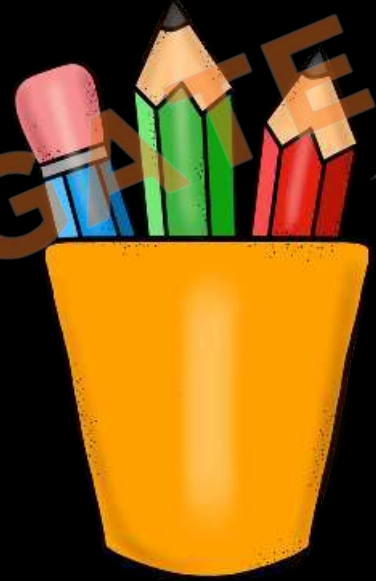


Defeats



Objects Revision

- 36. What is an Object
- 37. Objects Syntax
- 38. Accessing Objects
- 39. Inside Objects
- 40. Autoboxing
- 41. Object References
- 42. Object Shortcuts



Practice Exercise

Objects

1. Create **object** to represent a **product** from Myntra
2. Create an **Object** with **two references** and log changes to one object by **changing** the other one.
3. Use bracket notation to display **delivery-time**.
4. Given an object `{message: 'good job', status: 'complete'}`, use de-structuring to create two variables **message** and **status**.
5. Add **function** **isIdenticalProduct** to compare two product objects.



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JSON, Local Storage, Date & DOM

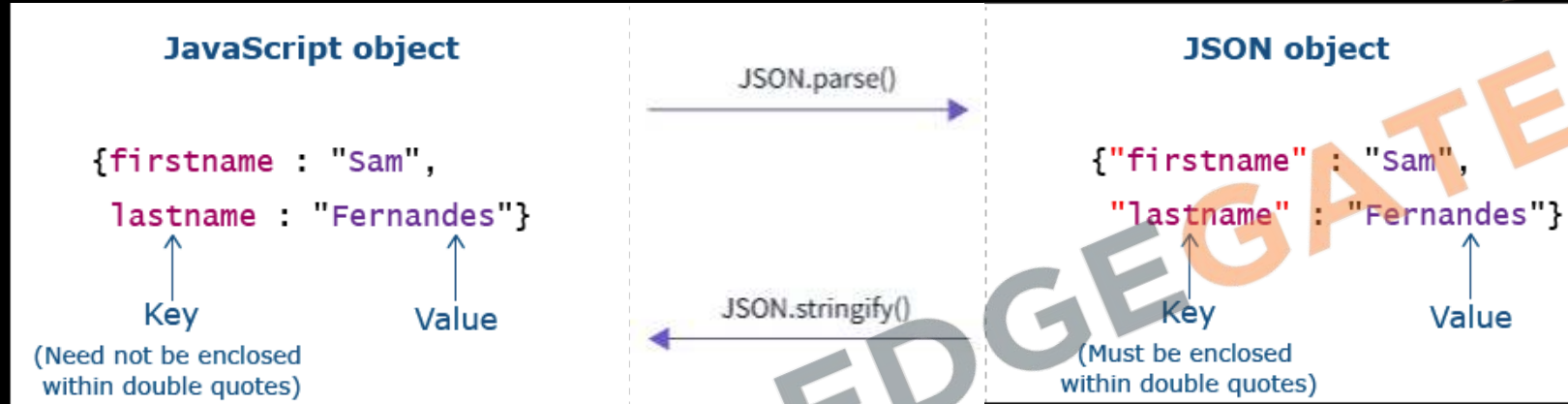
- 43. What is JSON?
- 44. Local Storage
- 45. Date
- 46. DOM Properties & Methods

Finish



Project

43. What is JSON?



1. **JavaScript Object Notation:** Not the same as JS object, but similar.
2. Common in network calls and data storage.
3. `JSON.stringify()` and `JSON.parse()`
4. Strings are easy to transport over network.
5. JSON requires double quotes. Escaped as `\`.
6. JSON is data format, JS object is a data structure.

44. Local Storage

```
// store an object in Local Storage
localStorage.setItem(
  "user",
  JSON.stringify({
    name: "Gopi Gorantala"
    age: 32
  });
);

// retrieve an object in Local Storage
const user = JSON.parse(
  localStorage.getItem("user")
);
```

Storage

- ▼ Local Storage
 - chrome://newtab/
- ▶ Session Storage
- IndexedDB
- Web SQL
- Cookies
- Trust Tokens
- Interest Groups



1. Persistent data storage in the browser.
2. `setItem`: Stores data as key-value pairs.
3. Only strings can be stored.
4. `getItem`: Retrieves data based on key.
5. Other Methods: `localStorage.clear()`, `removeItem()`.
6. Do not store sensitive information. Viewable in storage console.

Project Cricket Game



Defeats



1. Score will survive browser refresh.
2. Add Reset Button To clear or reset stored data.



Defeats



Defeats



45. Date



1. **new Date()** Creates a new Date object with the current date and time.
2. **Key Methods:**
 - **getTime():** Milliseconds since Epoch.
 - **getFullYear():** 4-digit year
 - **getDay():** Day of the week
 - **getMinutes():** Current minute
 - **getHours():** Current hour.
3. **Crucial** for timestamps, scheduling, etc.

46. DOM Properties & Methods

DOM and Element Properties

1. location
2. title
3. href
4. domain
5. innerHTML
6. innerText
7. classList

DOM and Element Methods

1. getElementById()
2. querySelector()
3. classList: add(), remove()
4. createElement()
5. appendChild()
6. removeChild()
7. replaceChild()

Project Cricket Game



Defeats



1. Show moves in the page instead of the alert
2. Show result in the page instead of the alert



Defeats



Defeats



Project Cricket Game



Defeats



1. Replace the Bat-Ball-Stump Buttons with Images



Defeats



Defeats



JSON, Local Storage, Date & DOM

Revision

- 43. What is JSON?
- 44. Local Storage
- 45. Date
- 46. DOM Properties & Methods



Practice Exercise

JSON ,Local Storage, Date & DOM

1. Display good morning, afternoon and night based on current hour.
2. Add the name to the output too.
3. Create a Button which shows the number how many times it has been pressed.
 - Also, it has different colors for when it has been pressed odd or even times.
 - The click count should also survive browser refresh.



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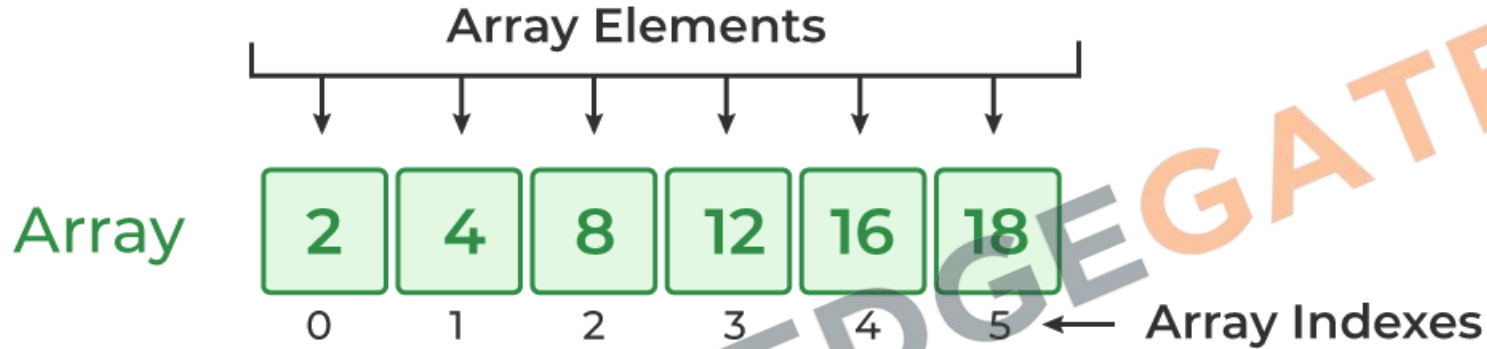


[Sanchit Socket](#)

Arrays & Loops

- 47. What is an Array?
- 48. Array Syntax & Values
- 49. Array Properties & Methods
- 50. What is a Loop?
- 51. While Loop
- 52. Do While Loop
- 53. For Loop
- 54. Accumulator Pattern
- 55. Break & Continue

47. What is an Array?



1. An Array is just a list of values.
2. Index: Starts with 0.
3. Arrays are used for storing multiple values in a single variable.

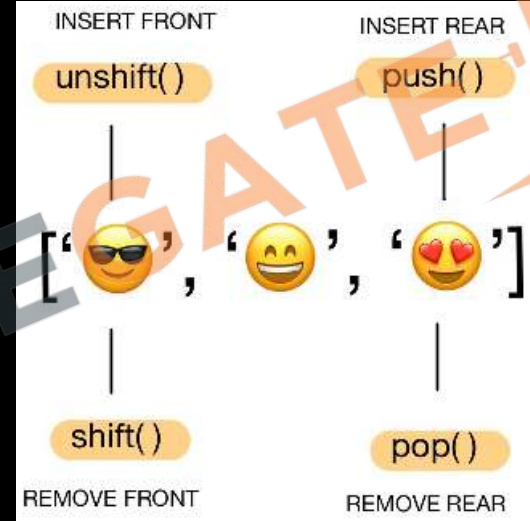
48. Array Syntax & Values

```
let myArray = [1, 'KG Coding', null, true,  
  {likes: '1 Million'}];
```

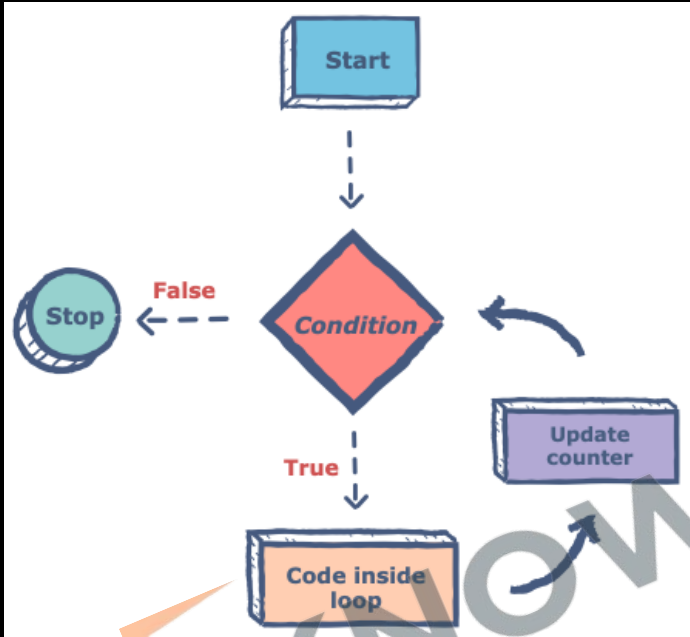
1. Use `[]` to create a new array, `[]` brackets enclose list of values
2. Arrays can be saved to a variable.
3. Accessing Values: Use `[]` with index.
4. Syntax Rules:
 - Brackets start and end the array.
 - Values separated by commas.
 - Can span multiple lines.
5. Arrays can hold any value, including arrays.
6. `typeof` operator on Array Returns Object.

49. Array Properties & Methods

1. `Array.isArray()` checks if a variable is an array.
2. `Length` property holds the size of the array.
3. Common Methods:
 - `push/pop`: Add or remove to end.
 - `shift/unshift`: Add or remove from front.
 - `splice`: Add or remove elements.
 - `toString`: Convert to string.
 - `sort`: Sort elements.
 - `valueOf`: Get array itself.
4. Arrays also use `reference` like objects.
5. De-structuring also `works` for Arrays.

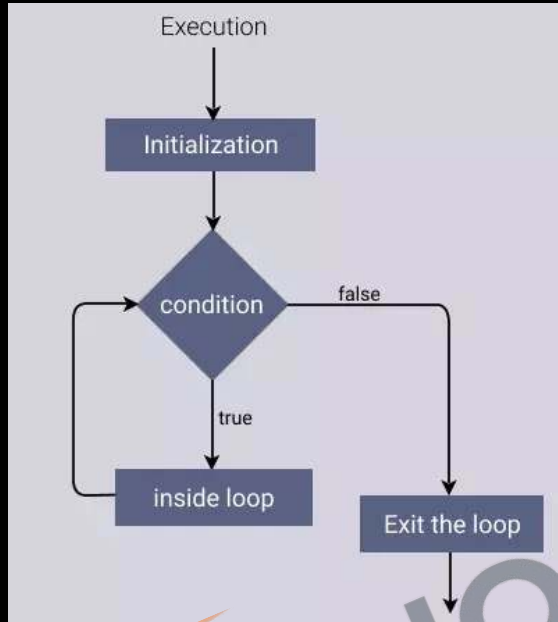


50. What is a Loop?



1. Code that runs multiple times based on a condition.
2. Loops also alter the flow of execution, similar to functions.
 - Functions: Reusable blocks of code.
 - Loops: Repeated execution of code.
3. Loops automate repetitive tasks.
4. Types of Loops: for, while, do-while.
5. Iterations: Number of times the loop runs.

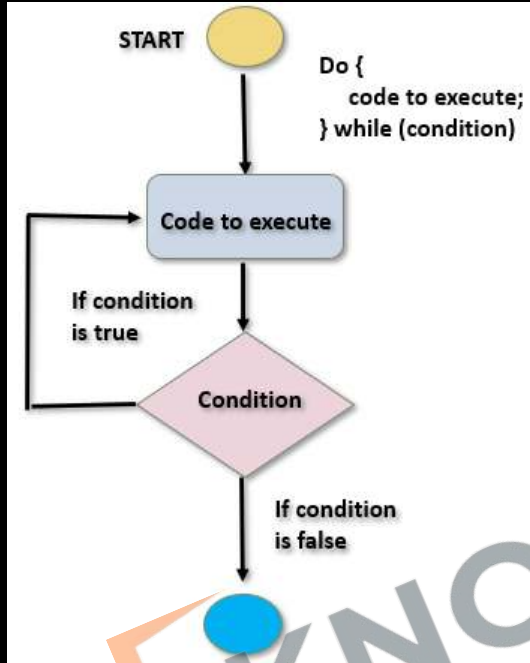
51. While Loop



```
while (condition) {  
    // Body of the loop  
}
```

1. **Iterations:** Number of times the loop runs.
2. **Used for** non-standard conditions.
3. **Repeating** a block of code while a condition is true.
4. **Remember:** Always include an update to avoid infinite loops.

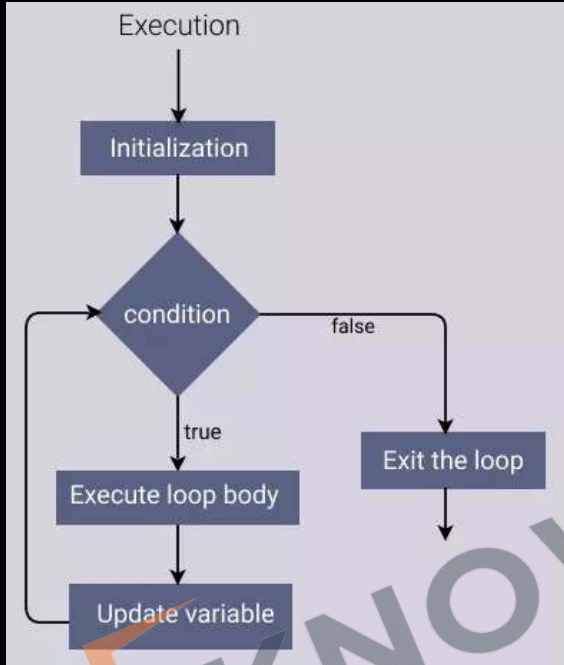
52.Do While Loop



```
do {  
    // Body of the loop  
}  
while (condition);
```

1. Executes block first, then checks condition.
2. Guaranteed to run at least one iteration.
3. Unlike while, first iteration is unconditional.
4. Don't forget to update condition to avoid infinite loops.

53. For Loop



```
for (initialisation; condition; update) {  
    // Body of the loop  
}
```

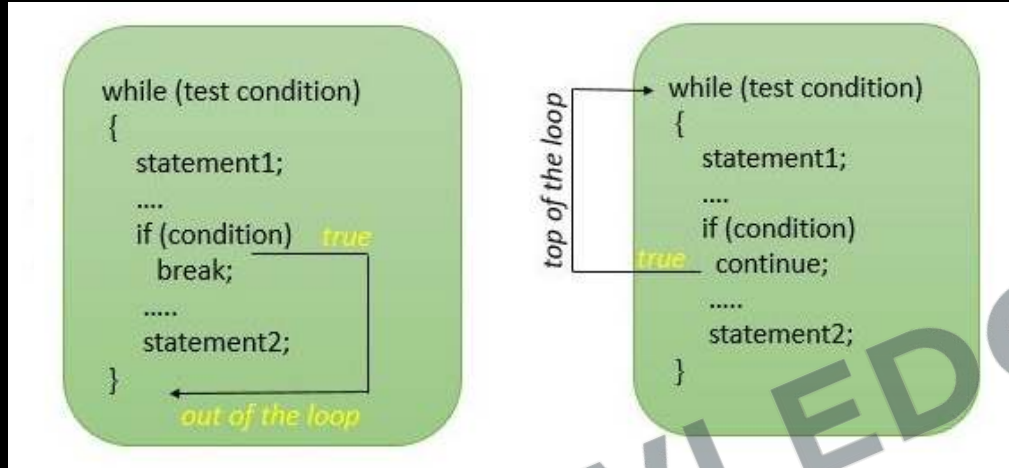
1. Standard loop for running code multiple times.
2. Generally preferred for counting iterations.

54. Accumulator Pattern



1. A pattern to accumulate values through looping.
2. Common Scenarios:
 - Sum all the numbers in an array.
 - Create a modified copy of an array.

55. Break & Continue

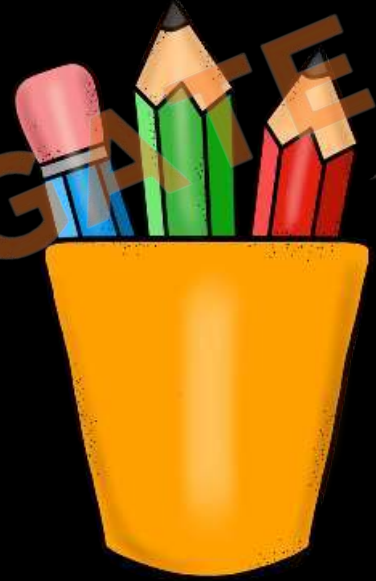


1. **Break** lets you stop a **loop** early, or **break out** of a loop
2. **Continue** is used to **skip one iteration** or the current iteration
3. In **while loop** remember to do the **increment manually** before using **continue**

Arrays & Loops

Revision

- 47. What is an Array?
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- 49. Array Properties & Methods
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- 53. For Loop
- 54. Accumulator Pattern
- 55. Break & Continue



Practice Exercise

Arrays & Loops

1. Create an array of numbers [5,6]. Add 4 at the beginning and 7 at the end of the array.
2. Create a method to return an element at a particular position in the array.
3. Create an array copy using slice method.
4. Create a while loop that exits after counting 5 prime numbers.
5. Modify the above loop to finish using break.
6. Create a for loop that prints number 1 to 10 in reverse order.
7. Using continue only print positive numbers from the given array [1,-6,5,7,-98]
8. Using accumulator pattern concatenate all the strings in the given array ['KG', 'Coding', 'Javascript', 'Course', 'is', 'Best']



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Advance Functions

- 56. Anonymous Functions As Values
- 57. Arrow functions
- 58. setTimeout & setInterval
- 59. Event Listener
- 60. For Each Loop
- 61. Array methods

56. Anonymous Functions As Values

```
let sum = function(num1, num2) {  
  return num1 + num2;  
}
```

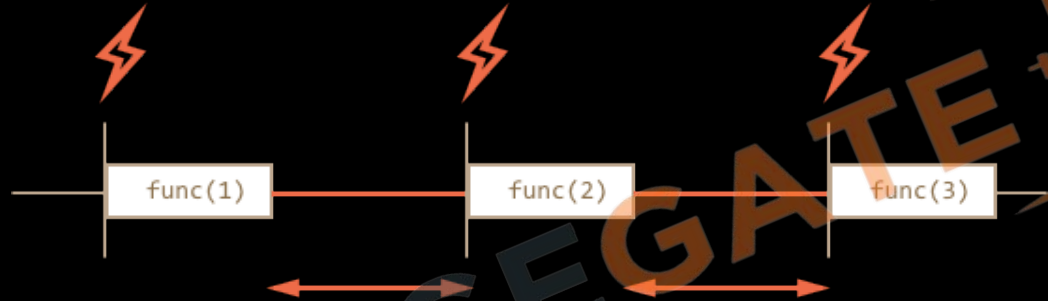
1. Functions in JavaScript are first-class citizens; they can be assigned to variables.
2. Functions defined without a name, often assigned to a variable.
3. Anonymous functions can be properties in objects
4. Can be passed as arguments to other functions.
5. Invoked using () after the function name or variable.
6. console.log(myFunction); and typeof myFunction will both indicate it's a function.

57. Arrow Functions

```
let sum = function(num1, num2) {  
  return num1 + num2;  
}  
  
let Sum1 = (num1, num2) => {  
  return num1 + num2;  
}  
  
let Sum2 = (num1, num2) => num1 + num2;  
let square = num => num * num;
```

1. A concise way to write anonymous functions.
2. For Single Argument: Round brackets optional.
3. For Single Line: Curly brackets and return optional.
4. Often used when passing functions as arguments.

58. setTimeout & setInterval



1. **Functions** for executing code **asynchronously** after a delay.
2. **setTimeout** runs **once**; **setInterval** runs **repeatedly**
3. **setTimeout**:
 - **Syntax**: `setTimeout(function, time)`
 - **Cancel**: `clearTimeout(timerID)`
4. **setInterval**:
 - **Syntax**: `setInterval(function, time)`
 - **Cancel**: `clearInterval(intervalID)`

59. Event Listener

```
const button = document.querySelector(".btn")
button.addEventListener("click", event => {
  console.log("Hello!");
})
```

1. What Is an Event: Occurrences like clicks, mouse movement, keyboard input (e.g., birthday, functions).
2. Using `querySelector` to attach listeners.
3. Multiple Listeners: You can add more than one.
4. `removeEventListener('event', functionVariable);`

60. For Each Loop

```
let foods = ['bread', 'rice', 'meat', 'pizza'];  
  
foods.forEach(function(food) {  
    console.log(food);  
})
```

1. A method for array iteration, often preferred for readability.
2. Parameters: One for item, optional second for index.
3. Using return is similar to continue in traditional loops.
4. Not straightforward to break out of a forEach loop.
5. When you need to perform an action on each array element and don't need to break early.

61. Array Methods

```
[🐮, 🥔, 🐔, 🌽].map(cook) => [🍔, 🍟, 🍗, 🍷]  
[🍔, 🍟, 🍗, 🍷].filter(isVegetarian) => [🍟, 🍷]
```

1. Filter Method:

- Syntax: `array.filter((value, index) => return true/false)`
- Use: **Filters** elements based on **condition**.

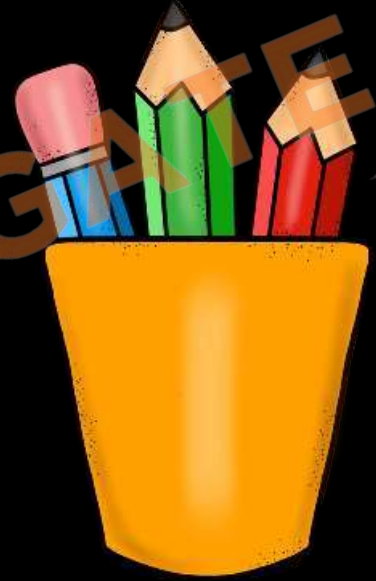
2. Map Method:

- Syntax: `array.map((value) => return newValue)`
- Use: **Transforms** each element.

Advance Functions

Revision

- 56. Anonymous Functions As Values
- 57. Arrow functions
- 58. setTimeout & setInterval
- 59. Event Listener
- 60. For Each Loop
- 61. Array methods



Practice Exercise

Advance Functions

1. Create a variable `multiply` and assign a function to this variable that `multiplies two numbers`. Call this method from the variable.
2. Create a function `runTwice` that takes a function as a parameter and then `runs that method` twice.
3. Create a button which should `grow` double in size on clicking after `2 seconds`.
4. In the `above exercise` add the behavior using an `event listener` instead of `onclick`.
5. Create a function that `sums` an array of numbers. Do this using a `for-each` loop.
6. Create a function that takes an `array of numbers` and return their `squares` using `map` function.



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